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SHOPLY APP

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ABSTRACT

Technological progress during the digitalization age changes retail commerce by creating user-specific and convenient shopping solutions. The Shoply app proposes to transform shopping practices by establishing a user-friendly platform which links customer demand to local businesses. The shopping application includes essential functions for browsing merchandise and placing orders followed by payment processing and immediate order tracking capabilities to enable continuous shopping convenience.

Shoply app exists to simplify the user experience by offering an efficient solution for customers to search and buy goods through their nearby stores. Local businesses obtain market expansion together with operational efficiency benefits from Shoply by combining various shopping features into a unified platform. Customers can experience shopping in a new way due to personal recommendations along with direct dialogue possibilities with shop vendors.

The application builds its core functionalities upon user privacy and security principles. The application safeguards user information with advanced encryption standards that follow current data protection standards. Through its use of present-day retail solutions and technological advancements, Shoply creates superior shopping satisfaction while promoting local merchants' success and better connects retail consumers with business sellers. Shoply has emerged as a strategic initiative which brings retail services closer to a unified and customer-oriented marketplace.

ACKNOWLEDGMENTS

We owe our deep appreciation to Dr. Abdullah for his insightful supervision and guidance throughout the Shoply app project development. Our supervisor Dr. Abdullah provided us with essential insights that supported us through all project stages to address problems and optimize the app's functionality according to our primary targets.

Our deep gratitude goes to our families who continually support us through their unlimited backing. The belief they have in our vision combined with their persistent motivation drove the team towards achieving this project's successful results.

Our appreciation goes to our friends who volunteered their time so we could obtain beneficial recommendations for the enhancement of our app. An essential role was provided by users who contributed directly to improved usability as well as complete user experience quality within the application.

Local business owners' merit special mention because they shared their experiences with us which enabled us to create a platform satisfying the needs of customers and shop owners. The practical and effective solution that Shoply offers to the local shopping community emerged from direct implementation of shop owner and customer feedback.

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Chapter 1

INTRODUCTION

Since the technological revolution shopping maintenance evolved from its conventional status as a daily necessity. The Shoply application serves as a comprehensive platform which combines local shops with their buyers for convenient shopping needs. Through this app users get to see products while making orders and reaching out to shop owners by using their mobile devices.

Shoply exists as a digital space to assist local businesses by enhancing their business operations through better management practices and expanding their reach to more customers. Through its platform Shoply enables customers to have an efficient personalized shopping journey. Shoppers can view merchandise selections and select stores they want to follow while receiving alerts for fresh products and bargaining opportunities. Both secure payment options and real-time order tracking are provided by the app while it offers straightforward buying procedures.

The direct link Shoply establishes between customers and sellers stands as the main distinctive factor of this marketplace model. Shop owners enable customers to maintain direct interactions through chat services so customers can obtain information and acquire recommendations thereby personalizing their shopping journey. Through their Shoply app shop owners can perform effortless product listing updates in addition to order management tasks which help boost their customer connections toward business expansion.

The Shoply application serves as more than an ordinary shopping platform since it acts as a gateway to connect nearby vendors with their target clientele. Local businesses achieve two benefits through Shoply because it offers both contemporary online shopping convenience with traditional shops familiarity enabling regional community support.

1.1 Problem Statement

Problem Statement: Small shops face enormous obstacles when trying to connect with contemporary consumers who lack online stores due to Bahrain's absence of a customized shopping platform. Inadequate customer connections and weak product exposure together lead to diminished sales opportunities. Online solutions available now focus on mega retail yet omit important elements such as personalization and live customer service that community-oriented shops require to thrive. Customer disappointment along with obstacles to business expansion and sustainability develop from this situation.

Scope of the Project: The main objective of this research and development initiative focuses on building the Shoply application platform that establishes direct store-customer interactions between local shop owners and their customers. Solo retailers can utilize the application for profile

management to list inventory and handle orders and sustain customer relationships in a social media style. Through the app users will get an interactive personal buying experience equipped with secure ordering capabilities alongside real-time shop interaction features. Business transparency as well as user satisfaction rates and the operational performance of participating shop owners will be assessed through the project's operations.

Research demonstrates that shopping customers now expect websites to provide personalized purchase experiences. Research conducted by Deloitte Digital (2022) reveals that personalization drives consumer purchasing behavior because 69% of people prefer to buy from companies offering tailored service. Businesses achieve better customer retention results through personalized interactions according to this survey (2022). Mobile apps that enable strong user-seller communication led to enhancing trust and user satisfaction in the market (Statista, 2023). Small businesses that lack essential features stay isolated from digital market opportunities while customers find it hard to locate and support their local brands in the market. The Shoply system fills these gaps through its program which enhances customer-business relations across Bahrain's market.

1.2 Project Objectives

Project Questions: -

1. What are the principal obstacles and gaps in current e-commerce systems which hinder local businesses from efficiently interacting with clients and competing in the digital landscape?
2. In what ways can a new retail-oriented mobile app be created to address the existing issues, serving as a one-stop platform for local shopping and business visibility?
3. What possible impacts can the introduction of an all-encompassing application such as Shoply have in promoting customer satisfaction, aiding small businesses, and supporting interaction between business proprietors and clientele?

Research Objectives: -

1. Evaluate the gaps of current e-commerce and retail applications based on the challenges of discoverability, absence of personalization, and poor interaction between the small businesses and clients.
2. We suggest the creation of a new digital platform (Shoply) which consolidates local stores and offers customized shopping services for each user while providing direct contact with store owners.
3. This research seeks to determine the impact of a consolidated shopping application on customer satisfaction, business visibility, and the growth of the local market through user feedback, data analysis, and benchmarking.

Application Objectives: -

1. A user-friendly interface should be developed to make sure customers and store owners can access easy browsing and ordering together with communication functions.

2. The system must establish a safe chat environment to let users communicate directly with nearby shopkeepers for asking questions obtaining guidance or receiving support.
3. The system includes an alert system to notify users when their followed shops make updates regarding products or orders or launch special promotions or state their current order status in real-time.
4. A security measure will ensure profile management functions operate securely by granting access only to shop owners and customers.
5. Through the dashboard system shop proprietors obtain complete control to handle their listings and monitor orders while providing customer responses and monitoring operational statistics.
6. Installation of a dependable scalable database framework for safe product user transaction information storage and retrieval will support system efficiency alongside complete data stability.

1.3 Project Significance

The entire project has great importance for the organization, individuals, and society for the following reason:

Organization:

- i. **Increased Competitiveness:** Implementing a fully digital platform dedicated to local business commerce will make the organization the leader in its field serving small enterprises with e-commerce needs. The specialized solution will position the organization ahead of conventional marketplaces because it directly meets unsatisfied needs of neighborhood businesses and community customers.
- ii. **Market Penetration:** The Shoply application targets users outside established markets through its solution of existing platform weakness in personalization and seller–customer interactions and visibility enhancement. User adoption expansion and multiple revenue channels together support business development and permit the scalability of offered services.
- iii. **Streamlined Processes:** The combination of various retail operations management procedures into one software application enhances workplace efficiency at all stages from product oversight to customer service to order monitoring to communicate Digital onboarding and business operation effectiveness will improve as the system reduces shop owner's operational workload while making local businesses function with less complexity and fewer resources.

Individuals:

- a. **Accessible and Convenient Shopping:** The integration of local shops into the Shoply application platform eliminates customer requirements to both search numerous scattered sources and physically visit retail locations. Users now benefit from one centralized digital

gateway to view and acquire goods from neighboring sellers during any time as they support contemporary shopping practices.

- b. **Personalized Experience:** Shoply enhances user experience with specialized product recommendations containing a shop-following system and location-dynamic suggestion capabilities. Through customization Shoply enables users to discover products which match their preferences, which produces both devoted customer relationships and improved shopping efficiency.
- c. **Time and Effort Savings:** A unified interface which handles product exploration and inquiries, orders and tracking together allows users to avoid multiple software or web tools for their needs. Thanks to this feature users save time in their daily purchasing activities which simultaneously speeds up navigation and task completion throughout the application.

Society:

- a. **Support for Local Economy:** Small independent business owners can join the digital market through the app without having to build expensive infrastructure or allocate marketing expenses. The local economy benefits from sustainable growth and economic sustainability through income circulation which occurs because of fair competition within this platform.
- b. **Knowledgeable Consumer Decision-Making:** Through these marketplace features including seller-seller interaction and present-time feedback and community assessment users gain better purchase insights. Sellers can feel confident with the visibility of direct communication that generates more transparency and strengthens customer trust overall.
- c. **Enhanced Commercial Efficiency:** The streamlined operations for transactions as well as communication and customer service at Shoply leads to enhancements in commercial efficiency. Better local retail service quality emerges when shop owners and customers enhance their communication flow and coordination through Shoply while experiencing shorter delays.

All things considered, this project is strategically significant because it will allow the organization to play a leading role in modernizing local commerce. It promotes economic inclusion and technological empowerment for small businesses in society while providing users with accessible, customized, and connected shopping experiences.

1.4 Report Outline

- This report's second chapter will use data from multiple sources to give a thorough overview of current retail and e-commerce applications, with an emphasis on how current platforms facilitate digital storefronts, customer interaction, and local business visibility.
- Project management will be covered in the third chapter, which will also highlight the process model that was chosen for the Shoply App project and provide evidence for its application. Along with discussing risk management strategies, this section will also describe the development activity plan.
- Data collection and analysis are the main topics of chapter four. It describes the research methods used, including user feedback analysis and surveys. The functional and non-

functional requirements required to construct the Shoply App platform are also presented in this chapter.

- The Shoply App system's user interface and design elements are examined in Chapter 5. The general system architecture, component-level design, and design principles utilized to create a seamless and user-friendly interface for both customers and store owners are covered in this section.
- The Shoply App implementation utilizes the described hardware and software tools which are presented in chapter six. The implementation process includes details about how essential system features operate in addition to descriptions of data handling systems and output reporting systems. The testing approaches together with the result evaluation method are discussed within this chapter.

Chapter 2

LITERATURE REVIEW

Academic research heavily relies on literature reviews because they evaluate existing knowledge as well as scholarly contributions throughout a specific field of study. The review establishes the research state-of-the-art then evaluates published work while exposing vacant areas and situates current research in its appropriate settings.

2.1 Related Systems

The following section reviews digital retail and e-commerce applications that unite sellers with buyers with special emphasis on systems designed for small and local businesses. The assessment covers a system-by-system review of functions alongside their advantages and shortcomings. An evaluation must determine the capabilities of existing solutions to enhance visibility and facilitate interaction and digital operation within small retail businesses.

By studying these platforms researchers can find essential weaknesses which consist of insufficient customizations and poor customer-seller connection capabilities and complicated interfaces that prevent small business owners from easy use. The analysis serves as groundwork to enhance the proposed Shoply App design by making it accessible, easy to use and efficient as a local commerce support platform.

2.1.1 Noon App (Noon E-commerce, Saudi Arabia, 2024)

The Noon app operates through a regional e-commerce network to unite customers and sellers under its platform that covers electronic items together with fashion products and home goods along with grocery listings. Through its application the company wants to improve online shopping by providing broad product choices along with simplified navigation and embedded payment features including delivery capabilities. Through its operations Noon has been instrumental to pushing forward online retail adoption across Saudi Arabia along with all GCC region states.

Analyzing Noon's role in digital commerce produces valuable information which can enhance the development of retail-oriented applications. A primary advantage of Noon's service is its capability to unite shopping experiences through letting multiple vendors sell products across one platform which defines optimal standards for multi-vendor digital systems. Noon proves that establishing centrally controlled scalable systems effectively serves retail domains that serve millions of users through daily transactions.

Shoply App follows the same basic principle of easy online shopping with vendor empowerment that Noon implemented as its platform design mainstay. An analysis of how Noon operates will

provide Shoply with essential knowledge about achieving better user experiences as well as improving communication paths and vendor-connectivity through studying its operational strengths and weaknesses particularly related to customization and seller-customer interaction. A system design featuring personalized options and community features will emerge from this analysis to develop Shoply into a marketplace that meets the requirements of local businesses.

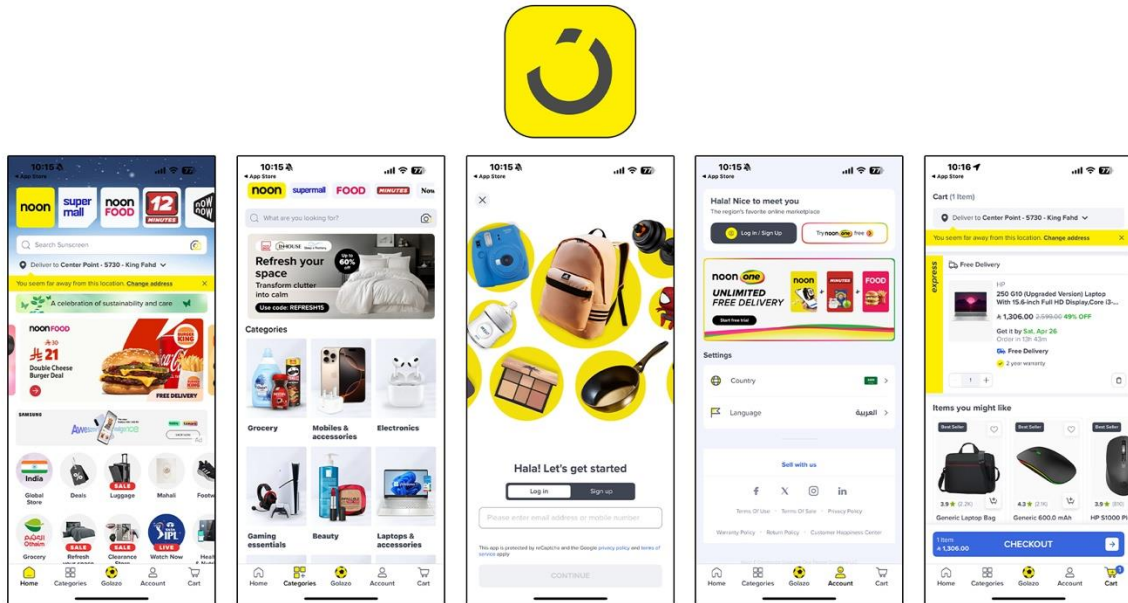


Figure 1: Noon App (Noon E-commerce, Saudi Arabia, 2024)

In Figure 1 shows how the Noon App (Noon E-commerce, Saudi Arabia, 2024) looks. The first picture depicts the main homepage where customers can see deals of the day, delivery modes, and offers by region. It provides immediate access to Noon's categories and services such as Noon Food and Supermarket. The second picture depicts the product categorized page where customers can browse through different product types such as groceries, electronics, and fashion. The third image offers the sign-up and sign-in page where users can sign up as new users with their phone number or email to begin shopping. The fourth image offers custom offers and offers access to Noon One, its free delivery membership program, country, and language settings. Finally, the fifth image is a product and checkout page with detailed information, where customers can view prices, delivery time, similar product suggestions, and proceed to order.

Advantages	Disadvantages
• Wide variety of products in multiple categories. • User-friendly interface for smooth navigation. • Integrated secure payment and delivery system. • Availability on both Android and iOS platforms.	• Limited support for personalized local shopping experience. • Minimal direct communication between sellers and buyers. • Smaller local sellers may struggle to compete on the platform. • Generic recommendations with limited customization.

Table 1: Advantages & Disadvantages - Noon App (Noon E-commerce, Saudi Arabia, 2024)

2.1.2 Dubizzle Bahrain App (formerly OLX Bahrain, 2012)

Users can access the Dubizzle Bahrain app after its reincarnation from OLX Bahrain in 2012 to enjoy online classifieds through a simplified platform. Dubizzle Bahrain's application enables Bahrain's people and small enterprises to purchase and offer numerous products and services between different categories. The application simplifies advertisement posting alongside user connection which leads to enhanced informal commerce accessibility in the region.

The Dubizzle platform allows free ad creation that combines photos, descriptions, pricing values and location information. Dubizzle provides both sellers and buyers with an easy-to-use platform which enables sellers to connect to various audiences while buyers can view items divided into electronics and furniture and fashion and real estate and cars sectors. Minimalist features alongside geographic search tools permit users to browse listings in an easy and fast manner.

The application features instant updates which allow users to instantly post or modify or remove listings. Sellers and buyers maintain direct communication channels within the app to deal with each other immediately through messaging or phone calls thus eliminating the requirement for third-party services. This operational model delivers flexible and rapid business operations and produces variable transaction results.

Through its featured listing function Dubizzle Bahrain allows sellers to promote their ads thus achieving better visibility alongside improved engagement from buyers. Users can locate available items through search tools and area-specific browsing which drives small-scale commerce in their local areas.

The platform fails to integrate payment features directly into the app but has proved popular for linking users who conduct street business thus becoming a leading marketplace in Bahrain. A comprehensive insight into developing Shoply App arises from studying the advantages and constraints of Dubizzle Bahrain. The developers can use Dubizzle's widespread adoption and simple approach to develop enhanced structured product listing features and secure payment systems along with an integrated shopping journey that serves Bahrain local businesses better for their users.

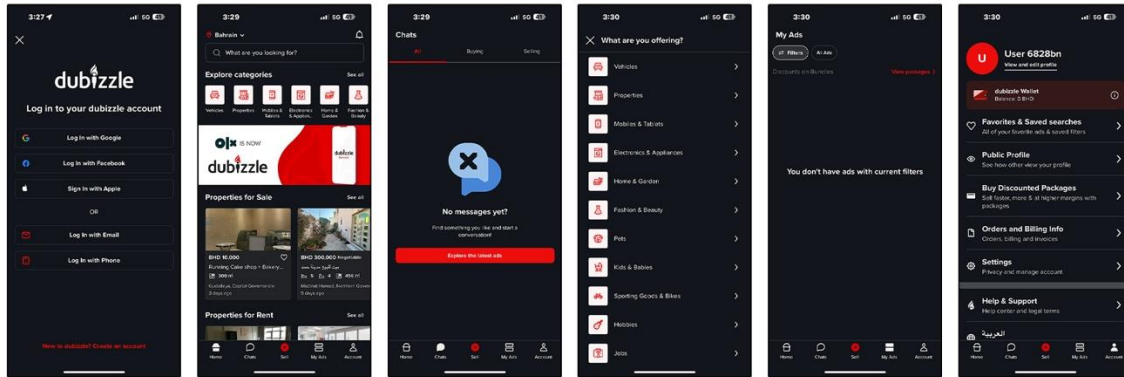


Figure 2: Dubizzle Bahrain App (formerly OLX Bahrain, 2012)

In Figure 2, we are presented with the user interface of the Dubizzle Bahrain app (Dubizzle, 2024). Users can navigate the app through a modern dark theme which offers quick functionality access during their browsing experience. The application interface provides logical pathways between categories as well as advertising sections and listing management features and user interaction capabilities. The main dashboard organizes its sections under "Home", "Sell", "My Ads", "Chats", and "Account" so users can easily transition between shopping and profile or listing management functions.

Users can authenticate their access through Google, Facebook, Apple ID or selecting manual login with telephone number or email. Users can explore different listing categories which include vehicles, electronics, furniture and real estate from the homepage and the system automatically applies location-based filters. A user-friendly system of icons leads buyers through the posting process to create new ads for sale.

The application provides users access for direct communication through its chat system and users can view profiles that include wallet features and save their favorite items while also offering filtering options for their personal advertisements. The well-designed layout combined with straightforward navigation structure enables easy accessibility for diverse users who want to make local sales rapidly.

Advantages	Disadvantages
• Easy to browse and navigate between categories. • Simple ads posting with minimal steps. • Accessible via iOS and Android platforms. • Suitable for local businesses and individuals alike.	• No built-in payment or checkout system. • Communication depends on external methods (calls/messages). • Lack of product filtering for professional stores. • Limited control over ad credibility and seller verification.

Table 2: Advantages & Disadvantages - Dubizzle Bahrain App (formerly OLX Bahrain, 2012)

Chapter 3

PROJECT MANAGEMENT

Project management consists of planning and executing projects through monitoring while closing projects for reaching predefined objectives which fall within pre-established scope and time along with budget constraints. A project's achievement depends on thorough planning together with resource distribution and ongoing quality standard checks. Time and cost together with quality form the fundamental pillars in project management because they establish the effectiveness and final success of the project.

3.1 Process Model

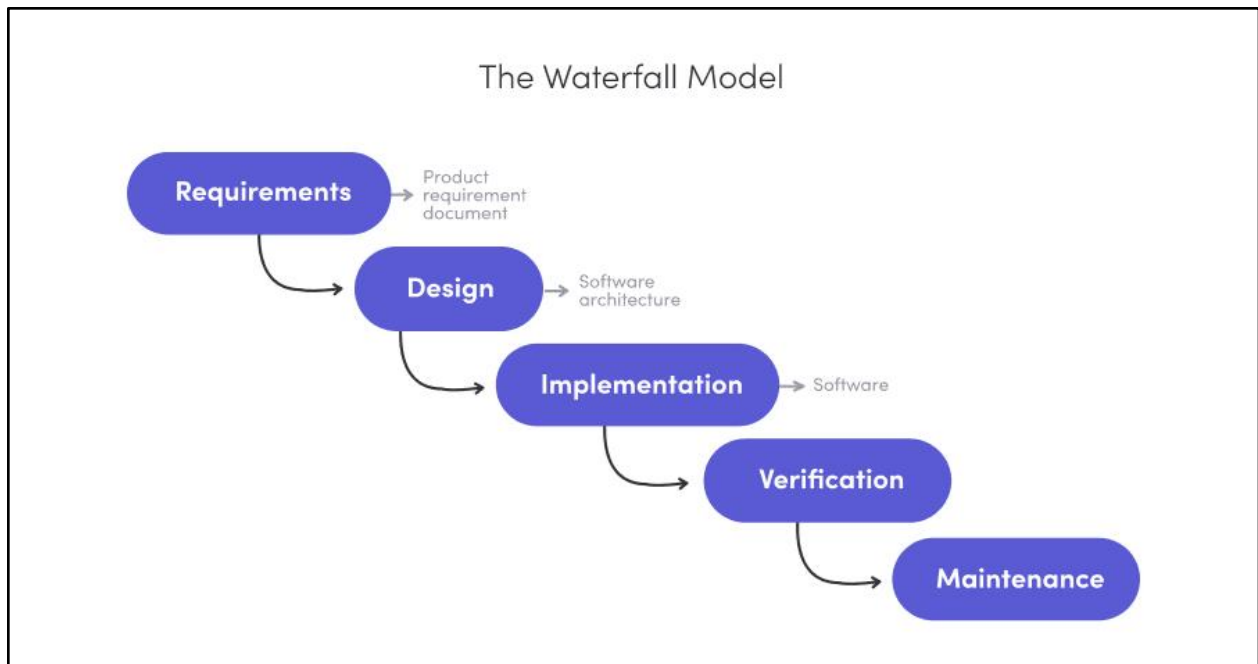


Figure 3: Process Model (monday blog Oct 21, 2024)

(monday blog Oct 21, 2024) The Waterfall Model exhibits a structured sequential project management strategy through Figure 3 because tasks follow a linear completion order. The model requires parts to finish their work before allowing advancement to the following procedure while

minimizing phase intersection. Such project planning method delivers organized workflows for projects having clear requirements alongside well-established specifications.

The Waterfall Model requires a project workflow that consists of requirements gathering then system design, then implementation followed by testing after which deployment occurs before maintenance begins. The method delivers high organizational control but shows strictness by interpreting all requirements beforehand resulting in reduced flexibility toward emerging project alterations. (monday blog Oct 21, 2024).

For our Shoply App, the Waterfall Model will be used, and it will consist of the following phases:

-

1. **Requirements Gathering:** The project team during this stage connects with stakeholders having local shop owners and prospective customers to obtain and document demands, and application features they expect. During this phase the platform's foundation will be built through the identification of essential needs which include shop profile setup alongside product posting functions along with customer shopping features and order handling capabilities.
2. **Design:** The following phase of system design begins after gathering all requirements. System development entails the project team designing an elaborate system layout which specifies application structure together with user interface and complete operational characteristics. The framework specified in this phase determines the functionality between shop listings along with posting mechanisms and order tracking features that fulfill all requirements.
3. **Implementation:** The development phase will execute the creation of the Shoply App. Following the design specifications the project team will create the shop's app features together with modules and back-end systems. Development work will include coding and integration of shop management tools, customer order tracking systems, payment gateways as well as testing for smooth access by shop owners alongside customers.
4. **Verification:** During implementation completion the software application will undergo entire testing and quality assurance examinations. The testing stage includes four main types of assessment: unit testing combined with integration testing followed by system testing and ending with user acceptance testing. Testing aims to find and resolve all platform bugs and errors together with usability issues before releasing the platform to deployment.
5. **Maintenance:** After deployment success the Shoply App will move to maintenance mode to deliver continuous support with updates. The application maintenance consists of checking app performance levels as well as resolving user-submitted problems and implementing new capabilities through business requirements and customer input. The application's security-system along with usability and operational effectiveness will benefit from continuing updates which enable adaptation to changing market requirements.

3.2 Project Risk Management

(Sana Commerce, 2022) The identification of any type of uncertainty or potential business damage in e-commerce counts as a risk factor. Digital marketplaces and other online platforms face different security hazards that decrease financial steadiness and operational performance. The identification of risks stands as an essential factor to guarantee business continuity because companies need structured risk management plans. To manage risks effectively businesses must first identify potential dangers and then rank threats to determine proper control procedures. (Sana Commerce, 2022).

3.2.1 Risk Identification

The identifier function of risk evaluation establishes the potential dangers which may impact online platforms. The Shoply App will evaluate potential threats throughout the development process and during implementation as well as throughout maintenance phases of the system. Technical risks for Shoply App consist of cyber vulnerabilities including data breaches alongside phishing attacks along with deceptive financial operations while operational risks embrace server breakdowns combined with payment breakdowns and administrative compliance tests. Business success depends on identifying these perils because effective risk management methods will protect platform security and reliability and extend its operational lifetime. (Sana Commerce, 2022).

Some potential risks to consider may include:

Table 3: Risk Identification (Sana Commerce, 2022)

No.	Risk Type	Possible Risk
1	Technology	System crashes or slow response times affecting shop visibility and customer engagement.
2	Data	Unauthorized access or data breaches compromising shop owners' and customers' information.
3	User	Shop owners may struggle to onboard and use the platform effectively, leading to reduced activity.
4	Operations	Delays in product updates or order processing due to system inefficiencies.
5	Competition	Difficulty in attracting users due to competition from larger e-commerce platforms. (Sana Commerce, 2022)

3.2.2 Risk Analysis

The process of risk analysis involves assessing recognized risks to determine their possible effects alongside probability which allows proper management strategy selection. Through qualitative and quantitative evaluation methods project teams identify risks to determine their potential outcomes and probabilities.

<i>Risk</i>	<i>Risk Probability</i>	<i>Risk Impact</i>
Shops leaving the platform due to low sales	Medium	High
Technical bugs affecting checkout process	High	Severe
Poor user engagement leading to low adoption	Medium	Serious
Data breaches compromising sensitive business information	High	Critical
Inconsistent order fulfillment causing customer dissatisfaction	Medium	Moderate
Lack of marketing reach limiting platform growth	High	Moderate

Table 4: Risk Identification (Sana Commerce, 2022)

3.3 Project Activities Plan

The shoply app project management is very challenging as it needs and wants very well-designed and organized coordination and arrangement from initial period of time (sana commerce 2022). For tasks in the project progress to be carried out smoothly, tasks inside the project need to be arranged effectively and structured for smooth workflow because confusion be avoided among the team members.

Correct task planning and scheduling as well as accurate estimation of the development timeframe is very important factors of successful project management. Having a clear roadmap in place will make operating smoother, teamwork go over more easily, and guarantee the project stays on track to get the job accomplished. (Sana Commerce, 2022).

Shoply application				
No.	Task Name	Duration	Start	Finish
1	Initiation	5 Days	02/13/2025	02/17/2025
1.1	Select a topic	1 Days	02/13/2025	02/14/2025
1.2	Search for topic	1 Days	02/13/2025	02/14/2025
1.3	Project Kickoff Meeting	1 Days	02/14/2025	02/15/2025
1.4	Determining the Project Manager and Supervisor	1 Days	02/14/2025	02/15/2025
1.5	Project objectives	1 Days	02/15/2025	02/16/2025
1.6	Project Significant	1 Days	02/16/2025	02/17/2025
1.7	Related System	1 Days	02/17/2025	02/18/2025
2	Planning	11 Days	02/20/2025	03/02/2025
2.1	Project Scope Management	5 Days	02/20/2025	02/24/2025
2.1.1	Work Break Structure Creation	3 Days	02/20/2025	02/22/2025
2.1.2	WBS dictionary	2 Days	02/22/2025	02/23/2025
2.2	Project Time Management	3 Days	02/24/2025	02/26/2025
2.2.1	Gantt Chart Activities	2 Days	02/25/2025	02/26/2025
2.3	Project Risk Management	2 Days	02/26/2025	02/27/2025

2.3.3	Risk Register	1 Days	02/27/2025	02/28/2025
3.1	Requirement Elicitation	3 Days	03/01/2025	03/03/2025
3.1.1	Survey Data Collection	2 Days	03/01/2025	03/02/2025
3.1.2	Survey Data Analysis	1 Days	03/03/2025	03/04/2025
3.2	System Requirements	12 Days	03/04/2025	03/15/2025
3.2.1	Functional Requirement	2 Days	03/04/2025	03/05/2025
3.2.2	Nonfunctional Requirement	2 Days	03/11/2025	03/12/2025
3.2.3	Requirement Traceability Matrix	3 Days	03/12/2025	03/14/2025
3.3	Personas	1 Days	03/16/2025	03/16/2025
3.4	System Models	3 Days	03/17/2025	03/19/2025
3.4.1	UML Diagram	2 Days	03/17/2025	03/18/2025
3.4.2	Process Specification	1 Days	03/18/2025	03/19/2025
4	Designing	11 Days	03/20/2025	03/31/2025
4.1	ADD iterations	11 Days	03/20/2025	03/31/2025
4.1.1	Review Inputs	1 Days	03/21/2025	03/21/2025
4.1.2	Establish The Iteration Goal By Selecting Drivers	1 Days	03/22/2025	03/22/2025
4.1.3	Choosing Reference Diagram	2 Days	03/22/2025	03/24/2025
4.1.4	Choose one or more elements of the system to refine	1 Days	03/25/2025	03/25/2025
4.1.5	Choose one or more design concepts	2 Days	03/26/2025	03/28/2025
4.1.6	Instantiate architectural elements	2 Days	03/28/2025	03/29/2025
4.1.7	Sketch views and record design decisions	1 Days	03/29/2025	03/29/2025
4.1.8	Perform analysis of the design	1 Days	03/30/2025	03/30/2025
4.2	Application Prototype	1 Days	03/31/2025	03/31/2025
5	Implementation	10 Days	04/04/2025	04/15/2025
5.1	Preparing tools and technical language	3 Days	04/12/2025	04/15/2025
5.2	Determining software (1)	5 Days	04/15/2025	04/20/2025
5.3	Determining software (2)	4 Days	04/21/2025	04/25/2025
6	Closing	5 Days	04/26/2025	04/30/2025
6.1	Combining the project	1 Days	05/01/2025	05/01/2025
6.2	Reviewing the work	1 Days	05/02/2025	05/02/2025
6.3	Specifying Future work	1 Days	05/03/2025	05/03/2025
	Total number of days	78 Days	02/13/2025	05/03/2025

Table 5: Gantt Chart

Our project's Gantt chart is provided in Figure 4. Between February 13th to February 17th, we took five days and targeted the initiation phase. In this, we selected the project topic and studied allied systems. From February 20th to March 2nd, we took part in the planning phase, and that lasted for 11 days. This involved creating all the project and system management diagrams and specifying functional as well as non-functional system requirements. Finally, from April 4th to May 3rd, we implemented and concluded the project.

Chapter 4

REQUIREMENT COLLECTION AND ANALYSIS

A description of the data collection process accompanies an analysis for creating the Shoply App which serves as a social media platform to link Bahrain shops without e-shops to local customers. An online survey served to collect required data about users' expectations and necessary needs.

4.1 Requirement Elicitation

During software development requirement elicitation stands as a fundamental procedure to obtain vital requirements which will enable system development. The online survey provided information about user preferences combined with shopping behavior and anticipated features of Shoply App.

4.1.1 Survey Questions and Analysis

- 1) What is your gender?
- 2) Which age group do you belong to?
- 3) How often do you shop from local businesses in Bahrain?
- 4) What are your main reasons for shopping from local businesses?
- 5) What improvements would encourage you to use an app like Shoply more frequently?
- 6) What features are most important to you in an online shopping app?
- 7) How comfortable are you with using mobile apps for online shopping?
- 8) Do you think a comprehensive shopping platform like Shoply is necessary in Bahrain?

4.1.2 Survey Analysis

The following report shares findings from our recent survey of both active local business shoppers and potential Shoply users based in Bahrain. An extensive group of 82 respondents provided their insights about consumer shopping behaviors throughout this data collection process. Feedback obtained from this survey helps Shoply develop its features because it ensures the platform serves local business owners and customers effectively in Bahrain's digital market.

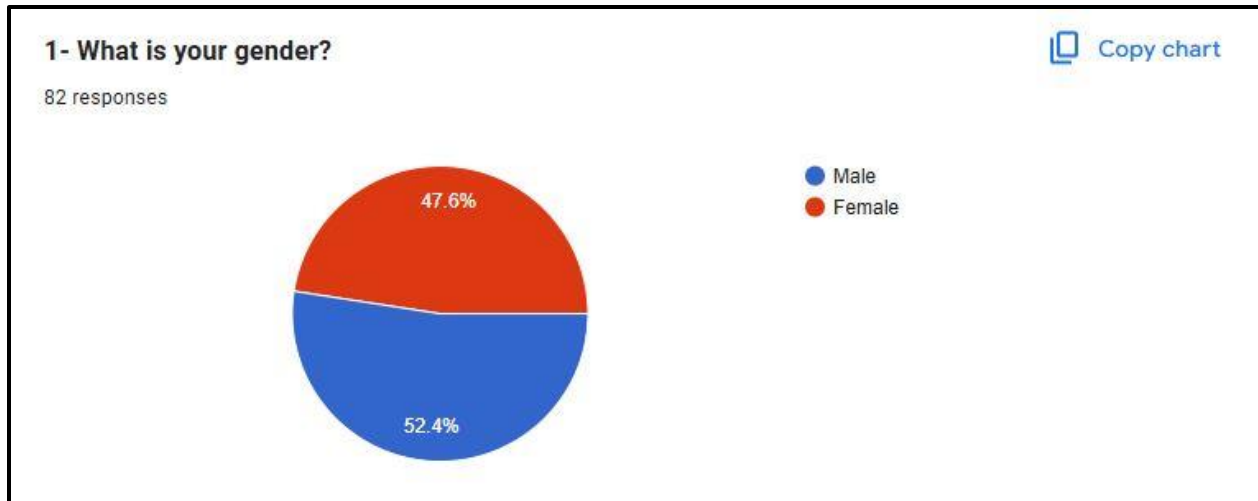


Figure 4: Survey Analysis

The survey pool received responses from 82 participants who divided into 52.4% males and 47.6% females. The balanced gender distribution indicates that Shoply needs to deliver equivalent service for male and female users to build an inclusive shopping platform. Such marketplace data supports marketing strategy development which highlights unisex characteristics and distinctive shopping tastes to diverse consumers.

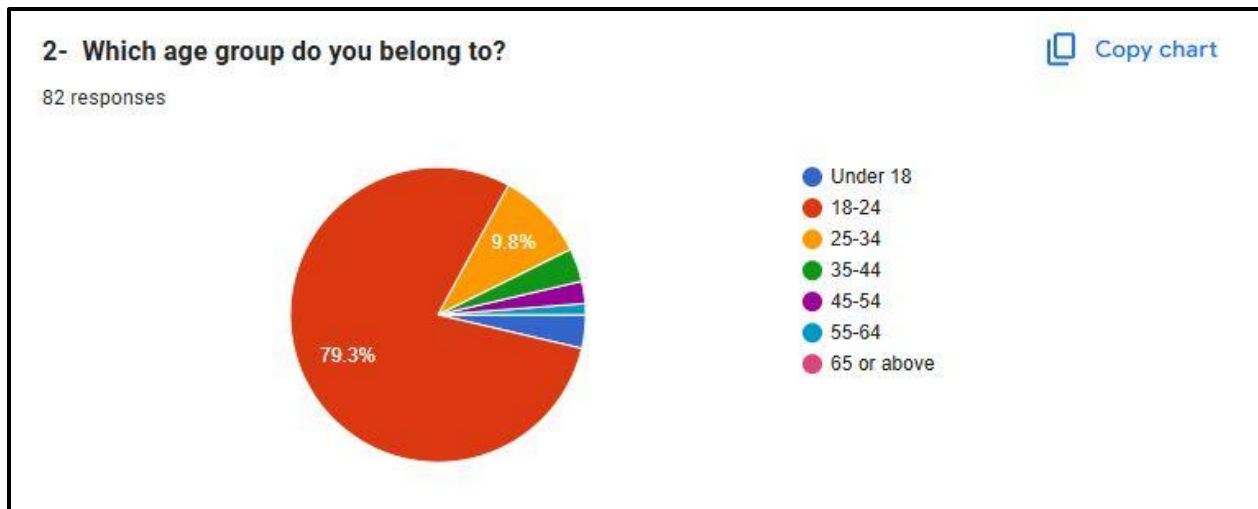


Figure 5: Survey Analysis

The population dominated by 18–24-year-olds forms 79.3% of the total sample group. The second largest demographic in this survey includes people between 25 and 34 years old who account for 9.8% of the total while other age groups have smaller representation. The respondents younger than 18 years old represent 3.7% of the population while those aged 35 years or older together comprise 7.3% of the survey participants. The research shows Shoply should focus its operations on young adults because they require innovative online shopping solutions which match their digital behavior patterns.

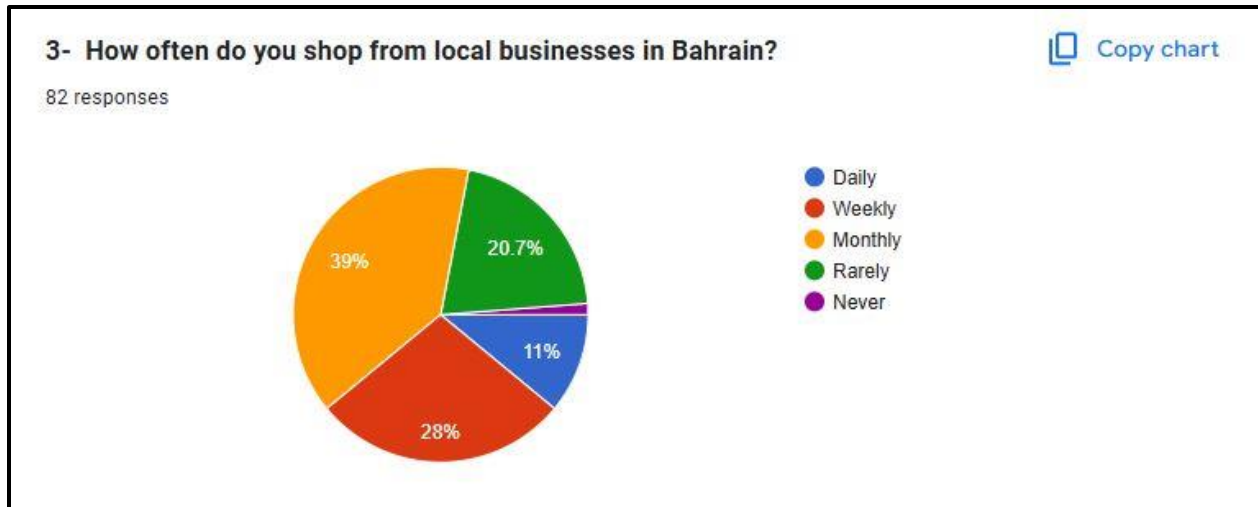


Figure 6: Survey Analysis

Survey data demonstrates that local business shopping occurs at a rate of 39% per month while 28% shop weekly. A significant segment of people (20.7%) avoids contact with local businesses while another segment (11%) shops every day. The data indicates that 1.2% of respondents never make any purchases from local stores in the community. Shoply should leverage these survey results to boost local business customer engagement through promotion strategies together with loyalty programs as well as convenient features for higher purchase frequency.

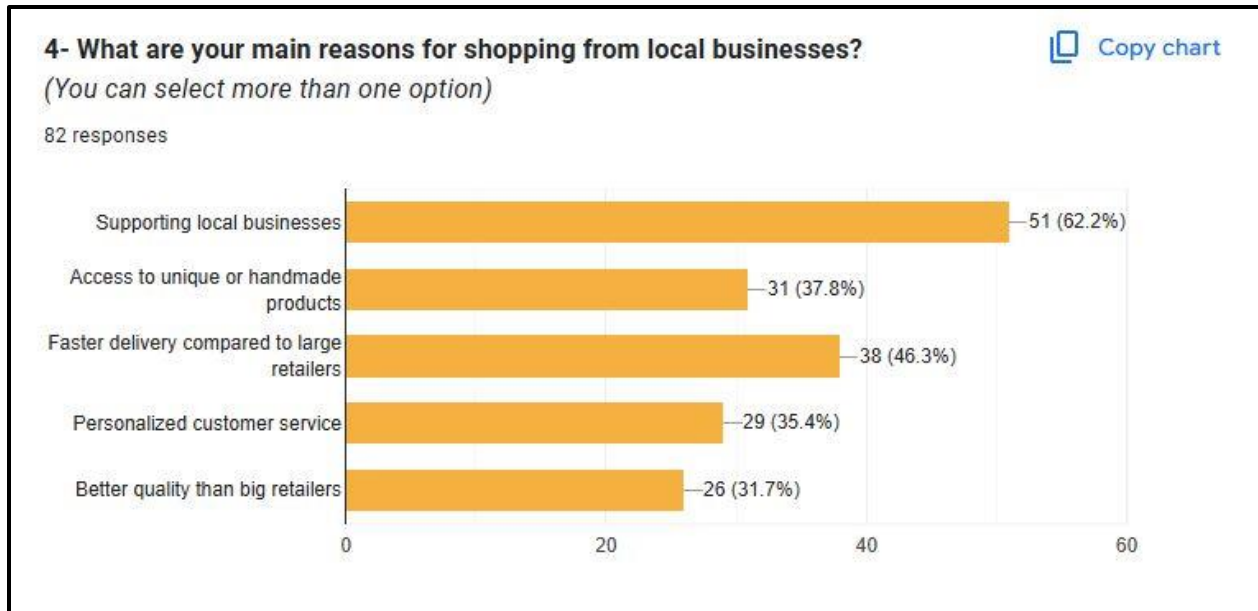


Figure 7: Survey Analysis

The key motivating force for supporting local businesses emerged from a survey of participants who numbered 62.2 percent. Users selected faster delivery features over big chains as the second major reason (46.3%) why they choose smaller shops. Additionally (37.8%), shoppers prefer unique handmade products in these local shops. People value personalized customer service from local shops at 35.4% along with preferring local products quality at 31.7% above large retailers. Research indicates Shoply should focus on showing customers how local products are unique and speeding up deliveries and improving personal connections for boosting consumer interaction.

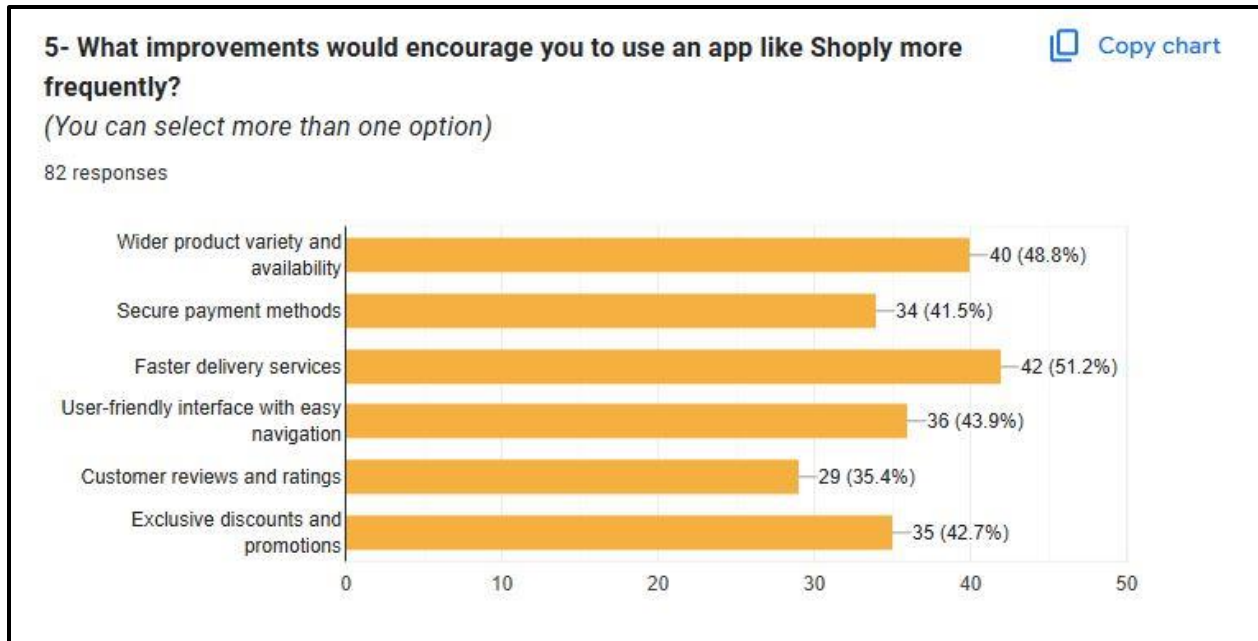


Figure 8: Survey Analysis

Most online shopping survey responders asked for faster delivery services which received 51.2% agreement from participants. The users place importance on both extensive product diversity and selection availability at 48.8%. Among the identified important factors were a user-friendly interface and easy navigation system at 43.9% while exclusive promotions and discounts stood at 42.7% and secure payment methods scored 41.5% and customer reviews and ratings reached 35.4%. The research findings demonstrate Shoply needs to focus on developing essential areas for improving user trust along with better applications usage frequency.

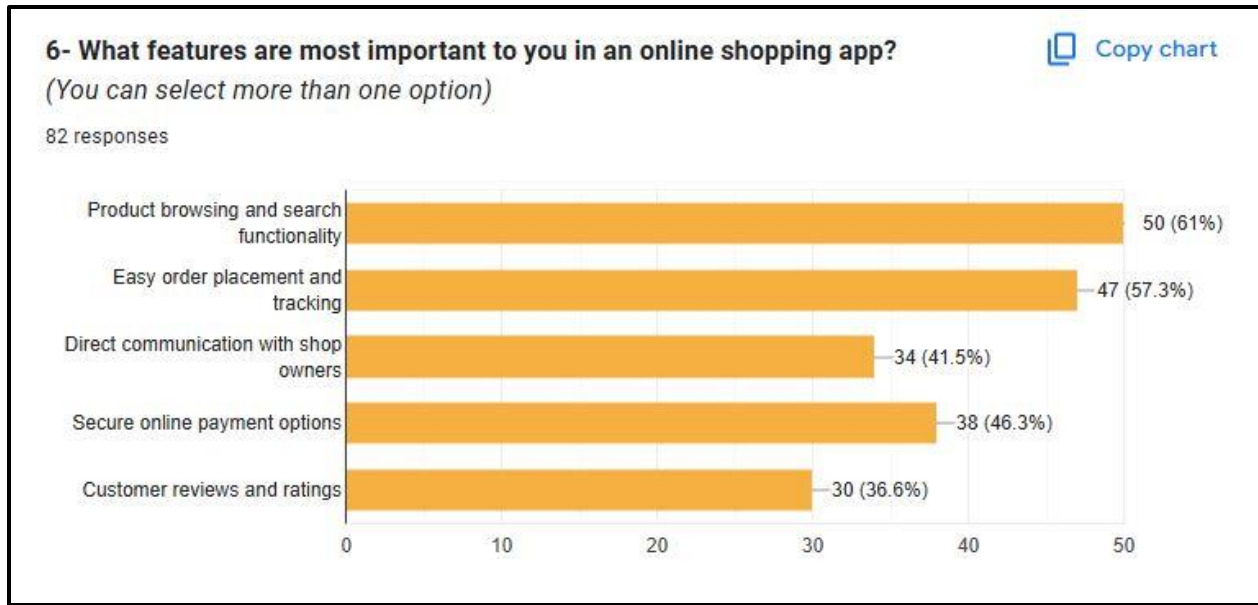


Figure 9: Survey Analysis

Users placed product browsing and search capabilities as their number one needed feature since it affected 61% of the respondents in positive ways. Order placement simplicity together with transparent tracking options stood at 57.3% as the second most important aspect (57.3%) to customers. For consumers secure payment options online along with direct communication opportunities with shop owners (46.3% and 41.5% respectively) were found to be critical factors representing the need for both transparent and convenient shopping methods. Customer reviews combined with ratings ranked as the third essential factor for respondents at 36.6 percent since they demonstrate the impact of social proof in purchasing behavior.

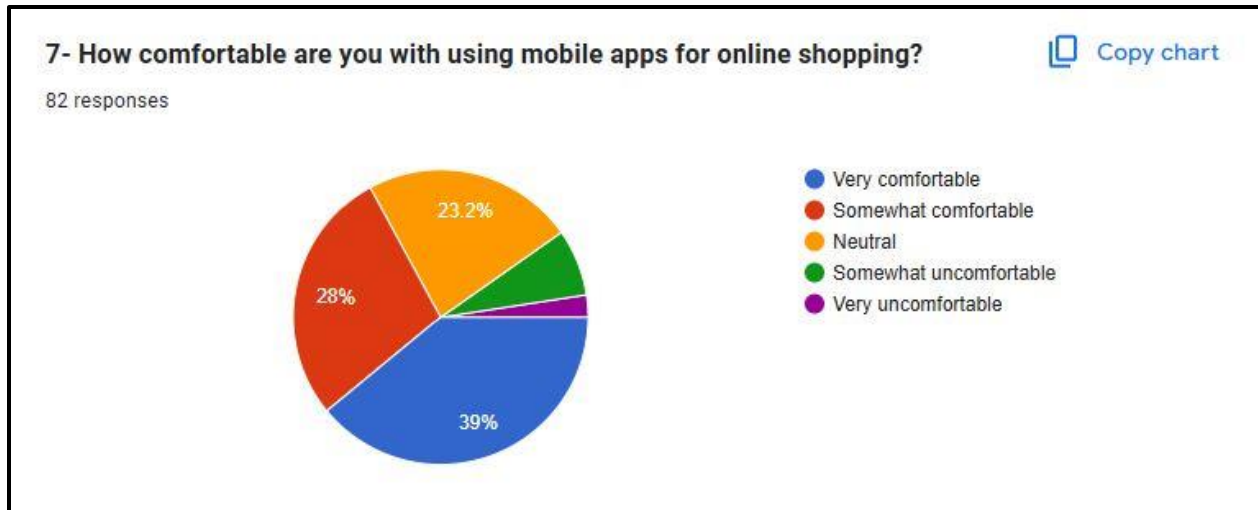


Figure 10: Survey Analysis

A substantial number of users (39%) reporting high comfort levels when using mobile shopping apps follow 28% expressing moderate comfort but 23.2% stayed neutral along with 7.3% and 2.4% who expressed somewhat uncomfortable to very uncomfortable feelings. About 23.2% of users show an ambivalent stance toward mobile shopping apps which indicates that certain users need reassurance and better app features to have a more satisfying shopping journey. Meager statistics show that 2.4% of all respondents expressed extreme discomfort with mobile shopping applications yet 7.3% shared moderate discomfort levels. The wide acceptance of mobile commerce stands alongside user comfort enhancement initiatives that lead to greater acceptance from hesitant users.

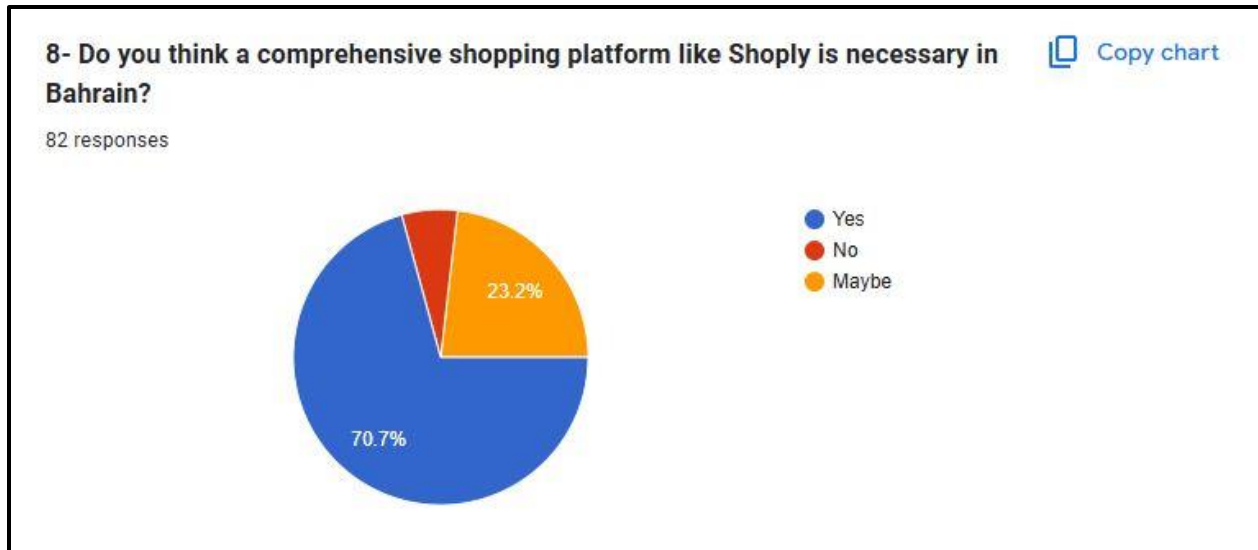


Figure 11: Survey Analysis

The survey revealed massive market demand for this type of platform since 70.7% of respondents argued for the necessity of Shoply in Bahrain. Further educational efforts need to be implemented because 23.2% of respondents expressed uncertainty about the Shoply platform's advantages. A total of 6.1% of participants believed that Shoply did not need to operate in Bahrain. Shoply demonstrates a strong market potential to become a fundamental business-consumer connection platform which will boost accessibility and economic development within Bahrain's markets.

4.2 System Functionality

System functionalities define the core capabilities and features offered by the Shoply application. Understanding these functionalities is crucial for ensuring that the system meets user expectations, performs efficiently, and provides a seamless shopping experience

User	User Registration and Authentication	Description: Users should be able to securely register and create profiles on the Shoply app.
		Acceptance Criteria: Users can sign up using valid personal details and authenticate their accounts through a secure login process.
	Login	Description: Users should be able to log in to their accounts using their email and password.
		Acceptance Criteria: Users can log in with valid credentials and access their profile and app features.
	Forget Password	Description: Users can reset their password using a verification code sent via email or mobile number.
		Acceptance Criteria: Users receive a reset link/code and successfully update their password
	Home Page	Description: Displays a feed of posts from followed shops and recommended products
		Acceptance Criteria: Posts are displayed in a scrollable format with options to like and save.

	Categories and Shop Discover	Description: Users can browse shops and products through categories.
		Acceptance Criteria: Products and shops are listed based on selected categories and search queries.
	Following Shops	Description: Users can follow their favorite shops to receive updates.
		Acceptance Criteria: Followed shops appear on the user's home page feed.
	My Orders	Description: Displays a list of all user orders, showing order status (Pending, Completed)
		Acceptance Criteria: Orders are updated in real-time with status changes.
	Order Details	Description: Users can view the details of each order, including product information and total cos
		Acceptance Criteria: Users can track order status and request additional support.
	Product Details	Description: Users can view detailed information about a product, including size, color, and price.
		Acceptance Criteria: Users can select a product variant and add it to the cart or wish list
	Cart	Description: Users can add products to their cart and modify quantitie
		Acceptance Criteria: Cart should update in real-time and display total cost.
	Home Page	Description: Users can complete their purchases using secure payment options
		Acceptance Criteria: Successful payment confirmation and order summary.
	Payment Methods	Description: Users can select from multiple payment options, including Apple Pay, Benefit Pay, and Cash on Delivery.
		Acceptance Criteria: Payment is processed securely and a confirmation is shown.
	Wish List	Description: Users can save products to a wish list for future reference
		Acceptance Criteria: Saved products are accessible from the user profile.

Admin	Account Management	Description: Admins can manage all user accounts
		Acceptance Criteria: Admins have the authority to add or remove user accounts and modify account details.
	Data Monitoring	Description: Admins can track app performance and user activity
		Acceptance Criteria: Reports on user engagement and system performance are available

Table 6: Functional Requirements

4.2.2 Non-Functional Requirements

Performance	Description: The system should load major pages within 3 seconds under normal network conditions.
	Acceptance Criteria: Pages load within 3 seconds during standard usage without delay.
Reliability and Availability	Description: The system must be available 24/7 with less than 1% downtime monthly.
	Acceptance Criteria: System uptime exceeds 99% monthly with minimal interruptions.
Security	Description: All data must be transmitted securely using HTTPS. User credentials must be encrypted.
	Acceptance Criteria: Users' login and sensitive data are protected and always encrypted.
Maintainability	Description: The code should follow Flutter best practices to allow easy updates and bug fixes.
	Acceptance Criteria: Future feature updates and bug fixes can be implemented without major app downtime
Scalability	Description: The system should support increasing numbers of users, shops, and products without performance degradation.
	Acceptance Criteria: The system handles up to 500 simultaneous users without performance issues.
	Description: The app should offer an intuitive user interface with a maximum of 3 clicks to find a product.

Usability	Acceptance Criteria: Users can easily navigate to desired products or shops within 3 steps.
Portability	Description: The app should function correctly on Android, iOS, and Web (future updates).
	Acceptance Criteria: Users can access the application seamlessly across multiple platforms.

Table 7: Nonfunctional Requirements

4.3 personas

4.3.1 Shop Owner

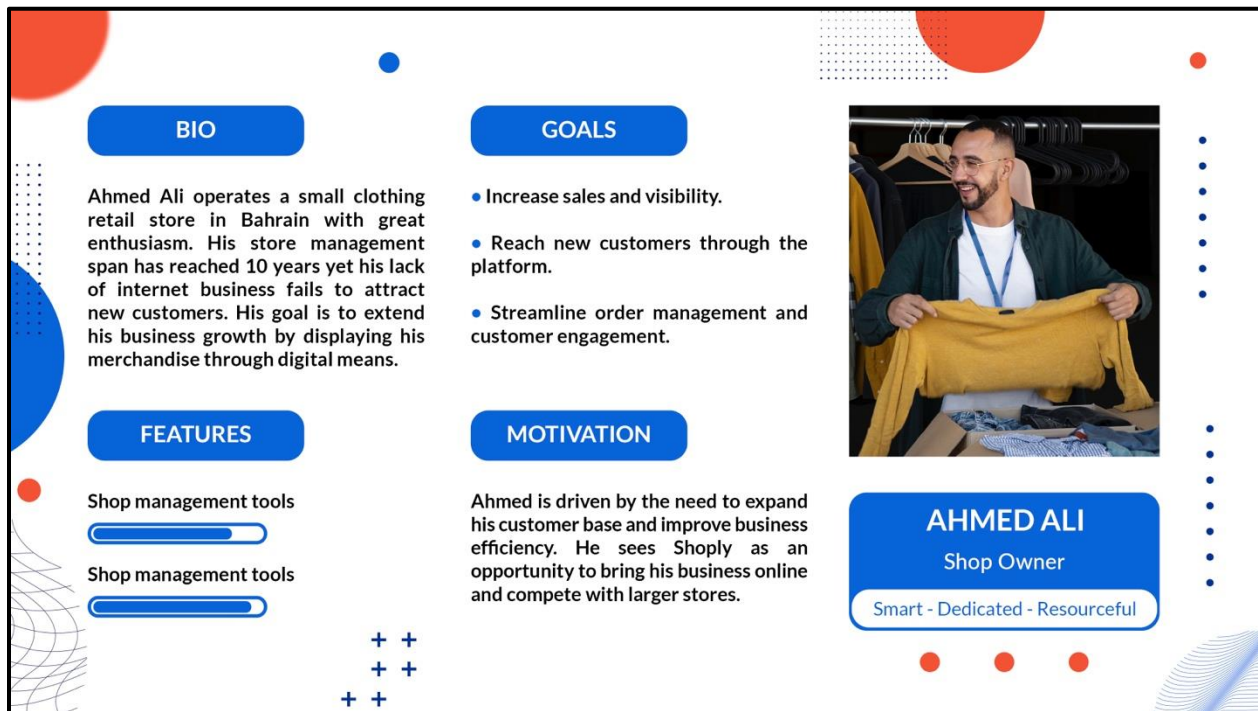


Figure 12: personas - Shop Owner

Scenario: Ahmed Ali accesses the Shoply App using his safe login credentials. Through his Shoply login Ahmed maintains his product inventory by posting new products while setting pricing details and stock availability. The application sends him a notification for a fresh order which he handles by working through the application dashboard. The customer satisfaction rating increases when Ahmed monitors reviews beside handling inquiries. Through Shoply's analytical tools Ahmed tracks business performance to determine pricing adjustments that will boost his sales numbers. The entrepreneur actively joins promotional activities which help boost product visibility and welcome new clients into his business.

4.3.2 Customer

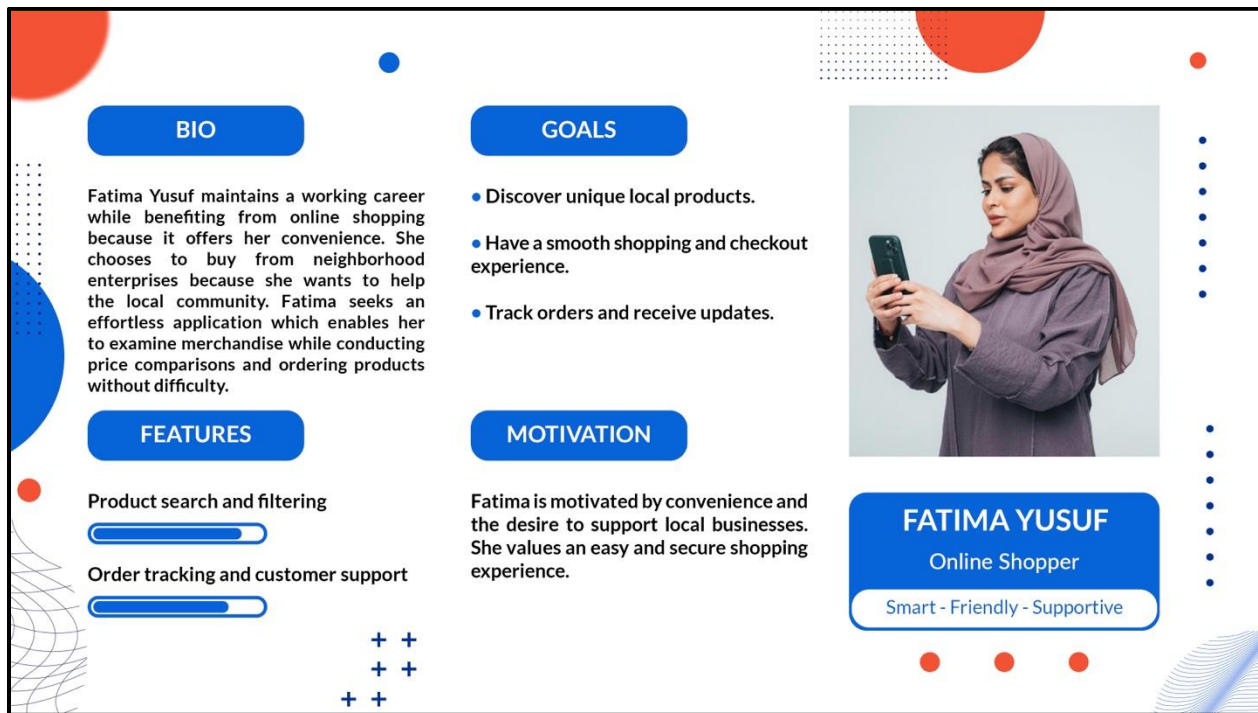


Figure 13: Personas - Customer

Scenario: Fatima Yusuf uses her phone to open Shoply App before exploring popular items in its available section. The boutique's handmade jewelry items attract her, so she begins their retail process by placing an order. The app both approves the payment and also displays predicted delivery schedules. Through the app Fatima checks the order status while receiving a notification when her purchased item gets shipped to her. She posts a positive review after receiving the item while starting to follow the boutique profile for future product and promotion notifications.

4.3.3 Admin

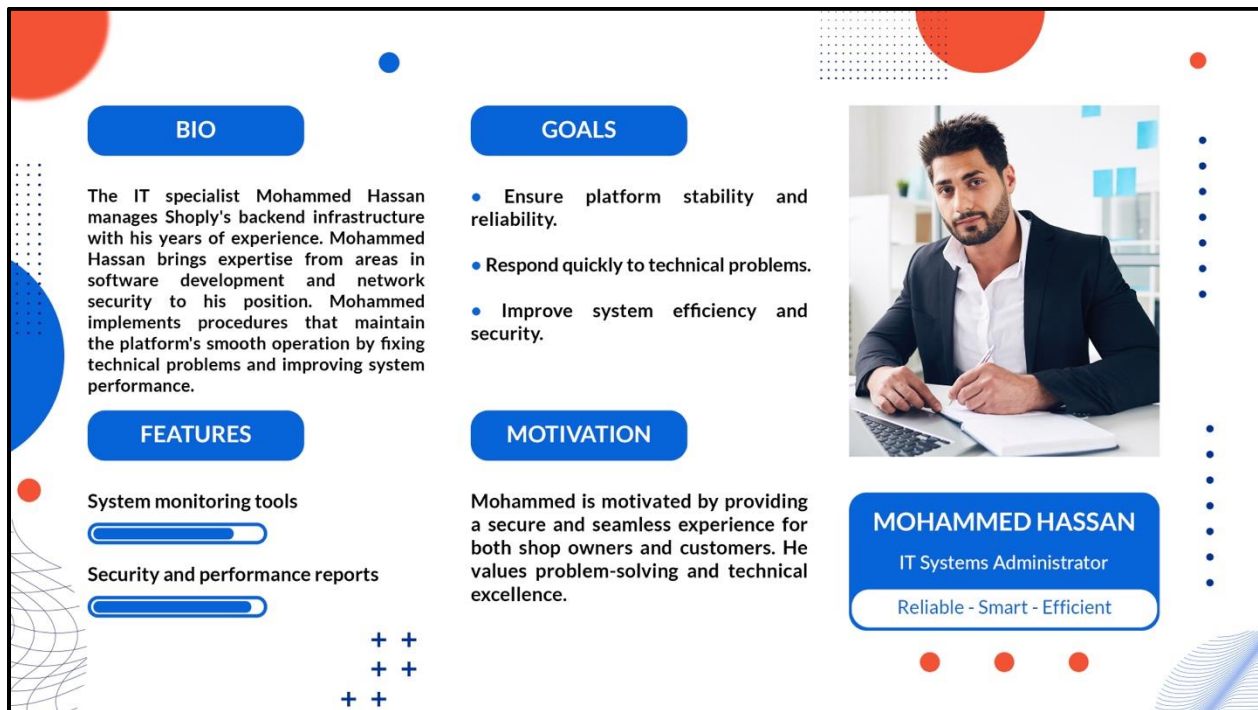


Figure 14: personas - Admin

Scenario: Mohammed accesses the administration interface of Shoply App to proceed with his tasks. The new shop applications undergo a review process by Mohammed before he verifies, they meet platform requirements to get approval. The IT partner checks system performance to address a small payment gateway problem that he solves right away. Through his role Mohammed conducts security updates on both encryption standards and data protection settings. Mixed feedback from shop owners and customers enables Mohammed to execute app functionality and user experience updates as part of his review process.

4.4 System Model

System models offer a way to see how data moves through the Shoply system and what kinds of information it collects.

Process elaboration in Shoply focuses on the user interactions with shops and products by choosing a subset of processes from the Data Flow Diagram (DFD) and expanding upon them to provide deeper insights.

The Entity-Relationship Diagram (ERD), which shows the relationships between users, shops, and products, is another essential feature of the system model.

4.4.1 Data Flow Diagram (DFD)

A data flow diagram (DFD) for Shoply shows how information flows through the platform between users, shops, and system databases. DFDs are used to evaluate how users register, browse products, and interact with shops within Shoply. They provide a clear visualization for both technical and non-technical audiences to understand how the app operates internally.

The four key components of Shoply's Data Flow Diagram are as follows:

1. **External Entity:** Represents external actors like users (customers) and shop owners. Users send data (such as registration information, product search queries) and receive data (such as product listings, shop details).
2. **Process:** Involves operations such as user registration, product browsing, adding products to cart, and placing orders. These processes handle user requests and generate outputs like shop recommendations or order confirmations.
3. **Data Store:** Includes Firebase Firestore where user profiles, shop information, product details, and order histories are stored securely. It ensures data is organized for efficient access and management.
4. **Data Flow:** Describes the movement of data between users, shops, and the system. For example, when a user signs up, their information flows to the database, and when they view a shop, product details flow from the database to the user's device.

In Figure 15 , the Data Flow Diagram (DFD) provides an overview of the system's architecture and data flow. It illustrates the input, processing, and output of data.

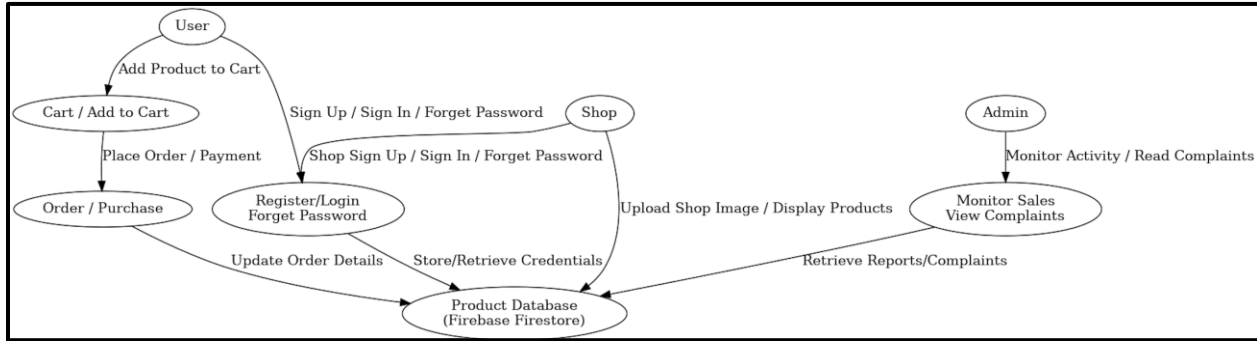


Figure 15: Data Flow Diagram (DFD)

in Figure 15, the Data Flow Diagram (DFD) is a system view that displays Shoply's architecture and data flow. It illustrates the interactions among the major entities such as User, Shop, Admin, and Product Database. The diagram depicts the data exchange and processing within the Shoply application. Stores and users are communicating with the system through actions such as Register/Login, Add to Cart, Buy, and Add Product Images. Arrows indicate the passage of information between these activities and entities, such as users signing up or logging in, adding products to the cart, purchasing, while the shop owners have control over displaying products, and admin monitor.

4.4.2 Process Specification

Process ID		1.0
Process Name	Authentication	
Process Description	Users can securely access the Shoply app by providing valid credentials (email and password). Authentication ensures that only authorized users can log in and access personalized features such as browsing shops and purchasing products.	
Input	User credentials (email and password)	
Output	Successfully login to the Shoply app and access the main user interface (home page).	
Logic	1. User enters email and password. 2. The app verifies the credentials against Firebase Authentication. 3. If credentials are valid, user gains access to the app's features. 4. Else, an error message (invalid credentials) is displayed.	

Table 8: Process Specification-Authentication

Process ID 2.0	
Process Name	User Registration
Process Description	New users can create a Shoply account by providing personal information including name, email, password, and user type (customer or shop owner).
Input	Full name, email, password, user type
Output	Account successfully created and stored in Firebase Firestore and Authentication.
Logic	1. User fills registration form. 2. The app validates the inputs. 3. If valid, account is created in Firebase Authentication and Firestore. 4. Else, display error messages (e.g., email already in use).

Table 9: Process Specification-User Registration

Process ID 3.0	
Process Name	Forget Password
Process Description	Users who forgot their password can reset it by receiving a verification code through their registered email.
Input	User's registered email address
Output	Password reset link/code sent to the user's email.
Logic	1. User submits registered email 2. System verifies email existence. 3. Send verification code/link. 4. User resets password using the code.

10 Table: Process Specification-Forget Password

Process ID 4.0	
Process Name	Add to Cart
Process Description	Users can add products from shop listings to their shopping cart for future purchase.
Input	Product details (ID, quantity)
Output	Product added to the user's cart stored in Firebase Firestore.
Logic	1. User clicks 'Add to Cart' on a product. 2. System stores product details in the user's cart in Firestore. 3. Update cart total if necessary.

11 Table: Process Specification- Add to Cart

Process ID		5.0
Process Name	Purchase	
Process Description	Users can complete their shopping by purchasing products in their cart using a payment method.	
Input	Cart items, delivery details, payment method	
Output	Order is placed and stored in Firebase Firestore; payment confirmed.	
Logic	1. User proceeds to checkout. 2. System verifies cart details. 3. User enters delivery and payment information. 4. Order is saved and payment is processed. 5. Confirmation message is displayed.	

12 Table: Process Specification-Purchase

4.4.3 Flowchart: Discussion tree

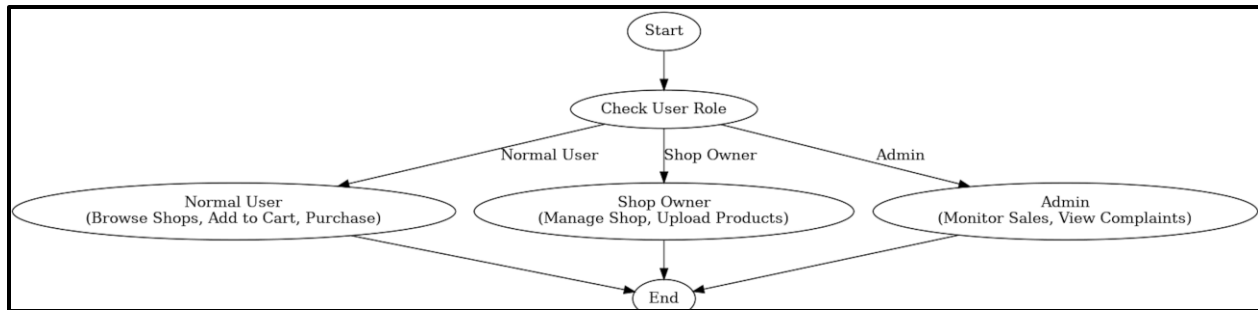


Figure 16: Flowchart: Discussion tree.

in Figure 16, the decision tree on how the Shoply system operates is illustrated in the flowchart. The user first logs into the application and the system verifies their role (normal user, shop owner, or admin).

If the user is a normal user, he/she can browse shops, view product listings, put products into carts, and place orders.

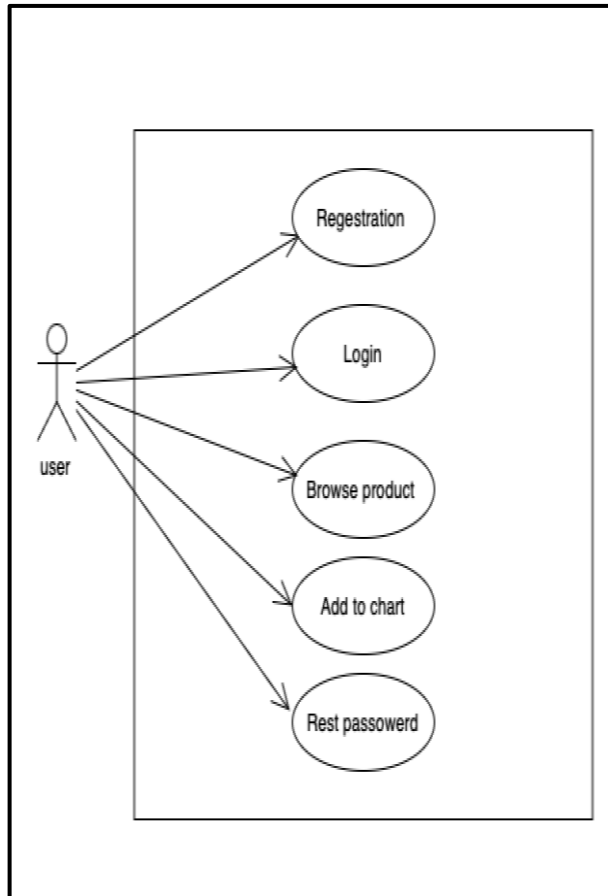
If the user is a store owner, he can edit his store profile, upload product photos, and customize product details offered to customers.

If the user happens to be an administrator, he can manage general sales activities, read users and shop complaints as well as suggestions, and maintain the running integrity of the system.

Arrows in the diagram show decision-making flow and actions based on the type of user who is accessing the system.

4.4.4 UML Use Case Diagram

User

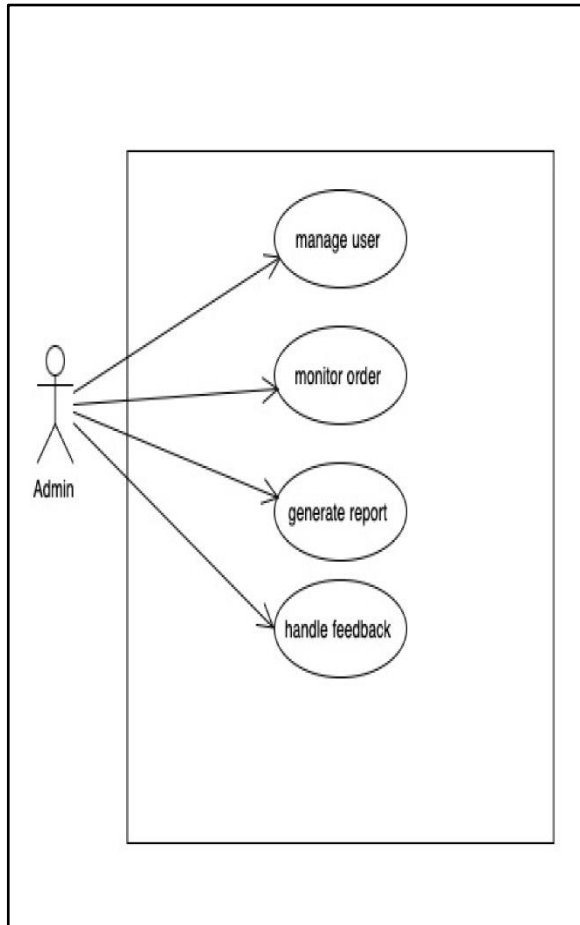


In Figure 19, the UML diagram fully captures the main elements and features available to ordinary users through the Shoply application. Some of these main features such as user registration, secure login, product browsing, search, managing wishlist, product addition to cart, ordering, payment, and tracking the order history are some of the major features.

The diagram also contains operations like giving feedback and resetting forgotten passwords. Serving as a good guide, this diagram visually depicts the user interface flow so that developers can apply an integrated and intuitive shopping process to Shoply consumers.

Figure 17: UML-User

Admin

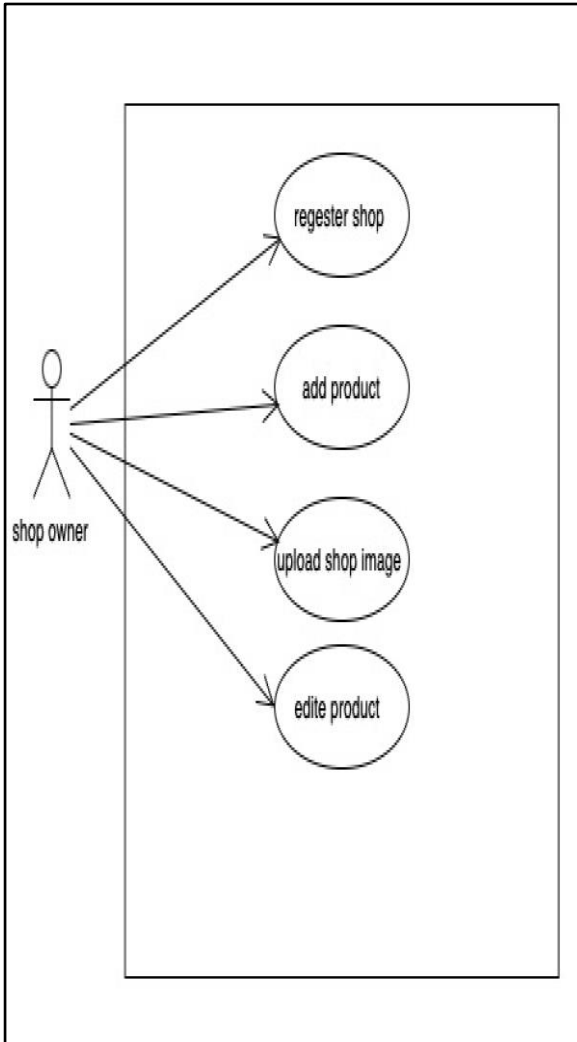


In Figure 20, the UML diagram elucidates the system administrator's primary functionalities and management controls systematically inside the Shoply application. Such features include safe login, administration of user accounts, moderating and viewing registered shops, feedback and complaint handling from customers, and observing order activities across the platform.

The diagram acts as a guide to backend control to enable administrators to effectively monitor and manage platform functioning. It provides an overview of the administrative interface, making management of the system straightforward, and emphasizes security, consistency, and platform integrity.

Figure18 : UML-Admin

Shop



In Figure 21, the UML diagram, shows the operation capabilities of shop owners in the Shoply application. It highlights key functionalities such as registration of shops, secure login, adding and managing products, editing product information, deleting items, viewing customer orders, updating order status, and uploading shop branding items such as banners and logos.

The diagram acts as a systematic blueprint for developers, assisting in the visualization of the shop owners' responsibilities and activities. It improves clarity in feature execution and helps develop an effective and user-friendly interface for business administration.

Figure 19: UML-Shop

4.4.5 UML Sequence Diagram [1]

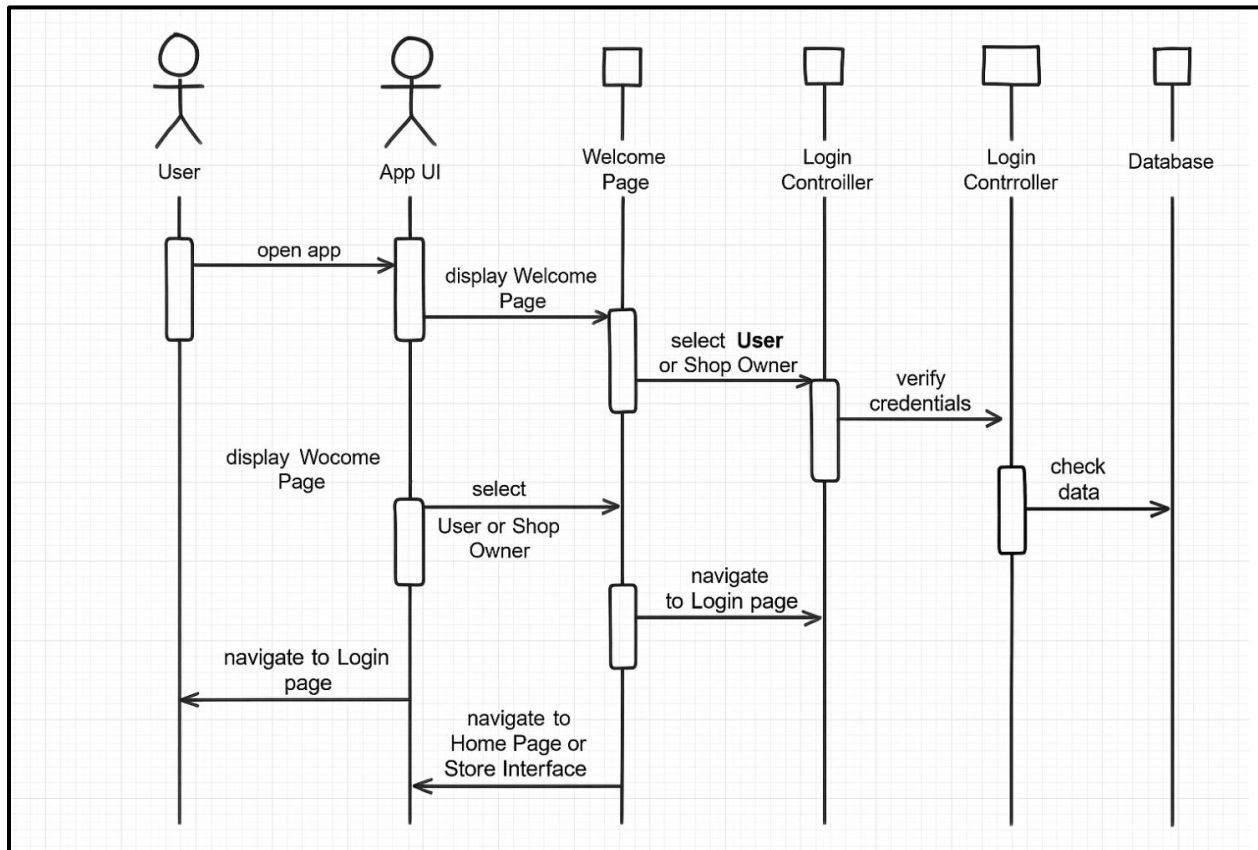


Figure 20: UML Sequence Diagram

In Figure X, we show the sequence diagram of the user login procedure in the Shoply app. The sequence diagram shows how the main parts, namely User, App UI, Welcome Page, Login Controller, and the Database, communicate with each other. The flow begins when the user starts the application, which begins with the App UI to display the Welcome Page. The user is then prompted to select whether he/she wants to start as a User or Shop Owner. After they are selected, the system prompts the user to the Login Page based on his/her role.

After they have entered their credentials, the Login Controller gets them and forwards them to the Database for authentication. Successfully authenticated, the app directs the user on to the correct interface, either the Home Page (for ordinary users) or the Store Interface (for owners).

This diagram depicts how user authentication is performed and maps out the sequence of decision logic concerning users.

4.4.6 UML Sequence Diagram [2]

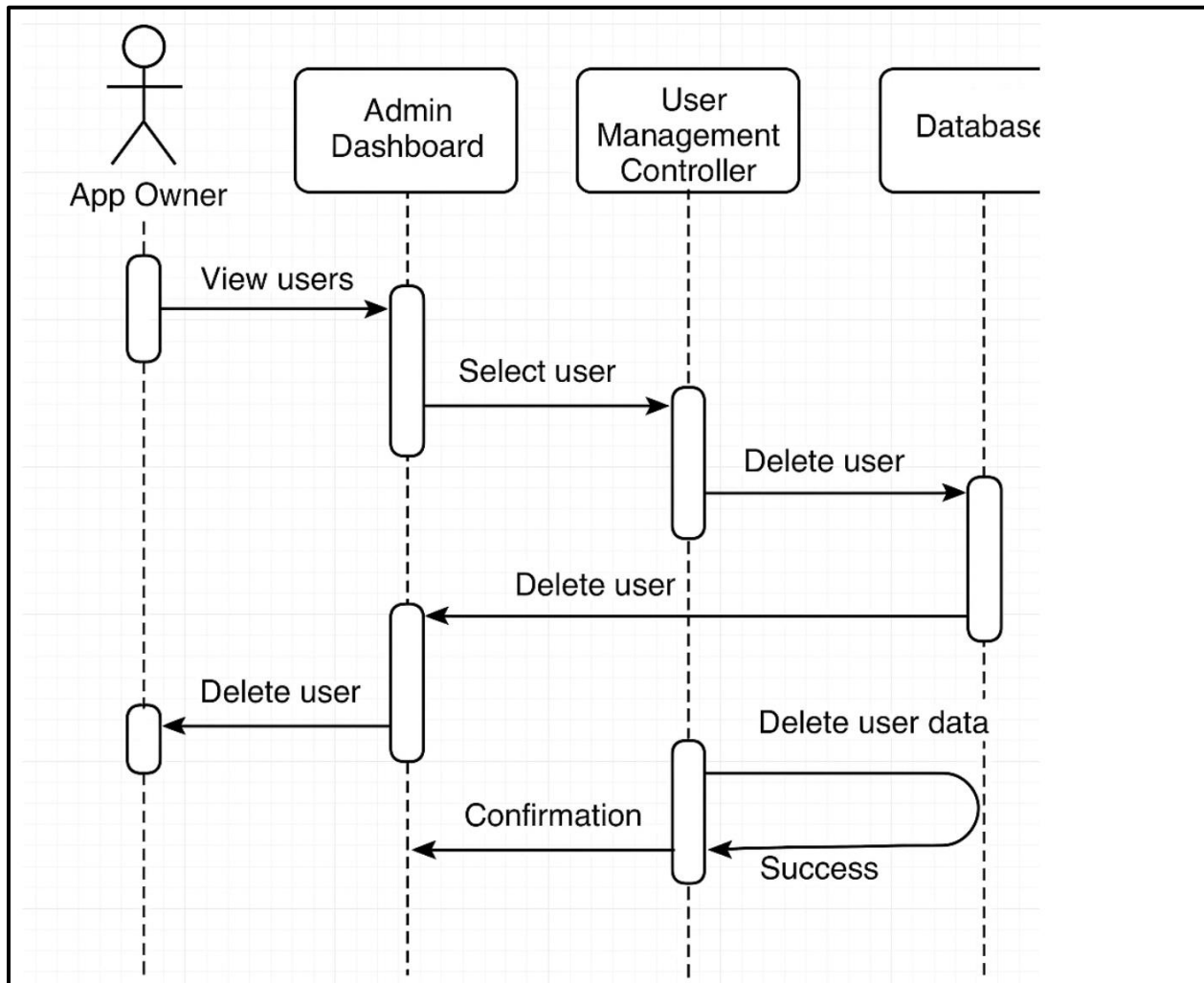


Figure 21: UML Sequence Diagram

In Figure X: illustrates a sequence diagram that indicates the communication between the App Owner, Admin Dashboard, User Management Controller, and Database within the Shoply App. The diagram clearly illustrates the process of user deletion from seeing the list of users, selecting a specific user, and sending a deleted request through the admin interface. The request is accepted by the management controller and completed by erasing the data of the user from the database. A reply message is given back to ascertain deletion.

4.4.7 UML Sequence Diagram [3]

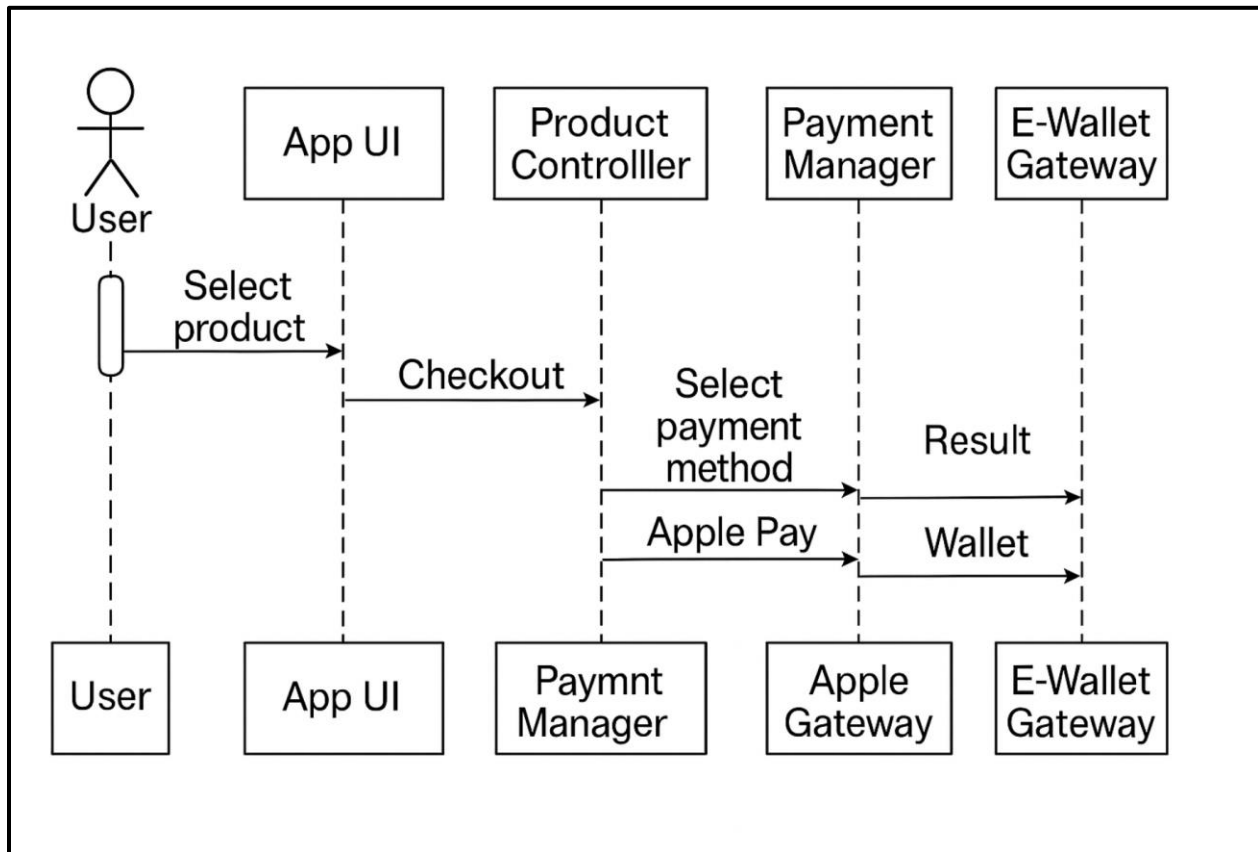


Figure 22: UML Sequence Diagram

in figure X: is a sequence diagram that describes the interaction between User, App UI, Product Controller, Payment Manager, and E-Wallet Gateway for checkout in the Shoply App. The flow is clearly depicted in the diagram, starting with product selection by the User and then the checkout. It then shows how the system handles the processing of payment method selection, communication with Apple Pay through the Payment Manager, and passes transaction results to the E-Wallet Gateway for completion.

4.4.8 Class Diagram

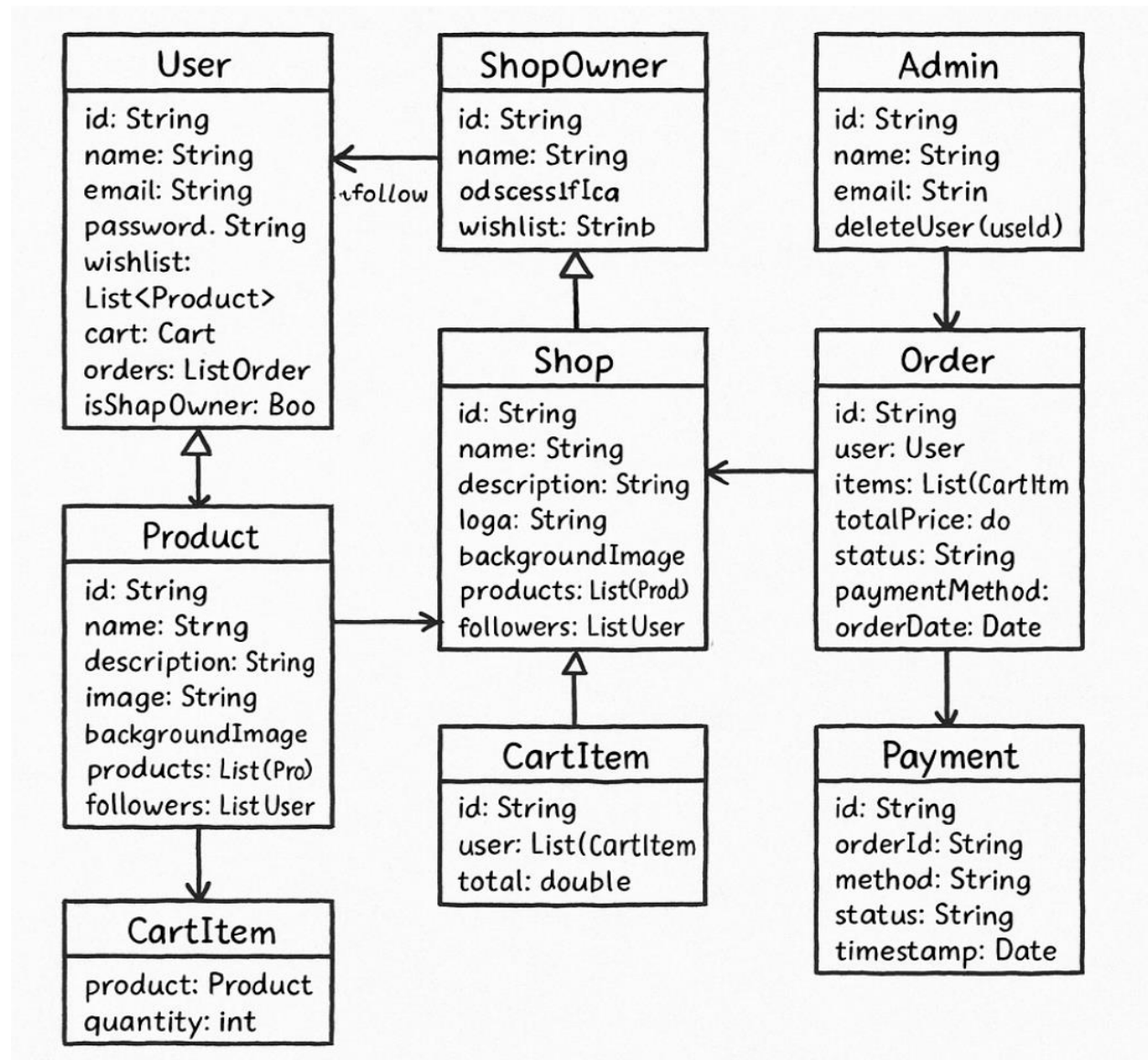


Figure 23: Class Diagram

In Figure X, we show the class diagram of our Shoply application. The diagram shows the main classes used in the application, such as User, ShopOwner, Admin, Shop, Product, CartItem, Order, and Payment. We can observe from the diagram how each of these classes is related to the others and what their attributes are. This gives an overall picture of the app's structure.

Since we developed the app with Firebase, and it's a flexible, non-relational database, the traditional Entity-Relationship Diagram (ERD) was not the most ideal for us. We chose to use a class diagram instead to model logic and data relationships in a better way. The diagram also enables developers to see the main features clearly and makes implementation easier.

Chapter 5

System Design

In this section, we will discuss the key components of our project, which include the database schema, procedure design, user interface design, system flow, and software algorithms.

5.1 Database Schema

We have designed a database schema for our project specifically dedicated to the Shoply application, which revolves around connecting users with local shops for a seamless shopping experience. The schema includes different collections like Users, Shops, Products, Orders, and Feedback.

The Users collection maintains information regarding app users, i.e., usernames, email IDs, passwords (handled securely through Firebase Authentication), profile images, phone numbers, account type whether customer or owner, and date of account creation.

The Shops table stores data about the shops operating under the platform. It has columns such as shop names, shop images, descriptions, shop owner user ID, and shop registration date. Each shop is linked to a user account so that ownership can be traced easily, and shops can be managed accordingly.

The Products list will store all the product information. There are fields such as product names, descriptions, prices, quantities, images, and shop ID for each. This enables every shop to update its product list independently and keep them updated in real time.

The Orders collection is used to track and control customer purchases in the application. It stores information such as the user ID, shop ID, list of products bought (with price and quantity), total price, order date, and payment status. This collection plays a pivotal role in aiding the purchasing process and maintaining records of complete orders.

Lastly, the Feedback table stores complaints and suggestions from users. It has columns such as the user ID, message content, type of message (complaint or suggestion), and date of submission. This facilitates administrators to monitor feedback on the site and implement required actions for improvement.

5.1.1 Users Table:

Field Name	Primary Key (PK)	Foreign Key (FK)	Data Type	Null
id	Yes	No	INT	No
fullName	No	No	STRING	No
email	No	No	STRING	No
passwordHash	No	No	STRING	No
profileImage	No	No	STRING (Optional)	Yes
role	No	No	STRING	No
createdAt	No	No	DATETIME	No

Table 13: Users Table

5.1.2 Shops Table:

Field Name	Primary Key (PK)	Foreign Key (FK)	Data Type	Null
id	Yes	No	INT	No
name	No	No	STRING	No
description	No	No	STRING	No
isFollowed	No	No	BOOLEAN	No
logo	No	No	STRING	No
backgroundImage	No	No	STRING	No

Table 14: Shops Table

5.1.3 Products Table:

Field Name	Primary Key (PK)	Foreign Key (FK)	Data Type	Null
id	Yes	No	INT	No
shop_id	No	Yes	INT	No
name	No	No	STRING	No
shortDescription	No	No	STRING	No
longDescription	No	No	STRING	No
quantity	No	No	INT	No
image	No	No	STRING	No
price	No	No	DECIMAL	No
category	No	No	STRING	No
isWishlisted	No	No	BOOLEAN	No
isInCart	No	No	BOOLEAN	No

Table 15: Products Table

5.1.4 Orders Table:

Field Name	Primary Key (PK)	Foreign Key (FK)	Data Type	Null
id	Yes	No	INT	No
serialNumber	No	No	STRING	No
status	No	No	STRING	No
date	No	No	DATETIME	No

Table 16: Orders Table

5.1.5 Order Items Table:

Field Name	Primary Key (PK)	Foreign Key (FK)	Data Type	Null
id	Yes	No	INT	No
order_id	No	Yes	INT	No
name	No	No	STRING	No
shortDescription	No	No	STRING	No
longDescription	No	No	STRING	No
quantity	No	No	INT	No
image	No	No	STRING	No
price	No	No	DECIMAL	No

Table 17: Order Items Table

5.1.6 E-R diagram:

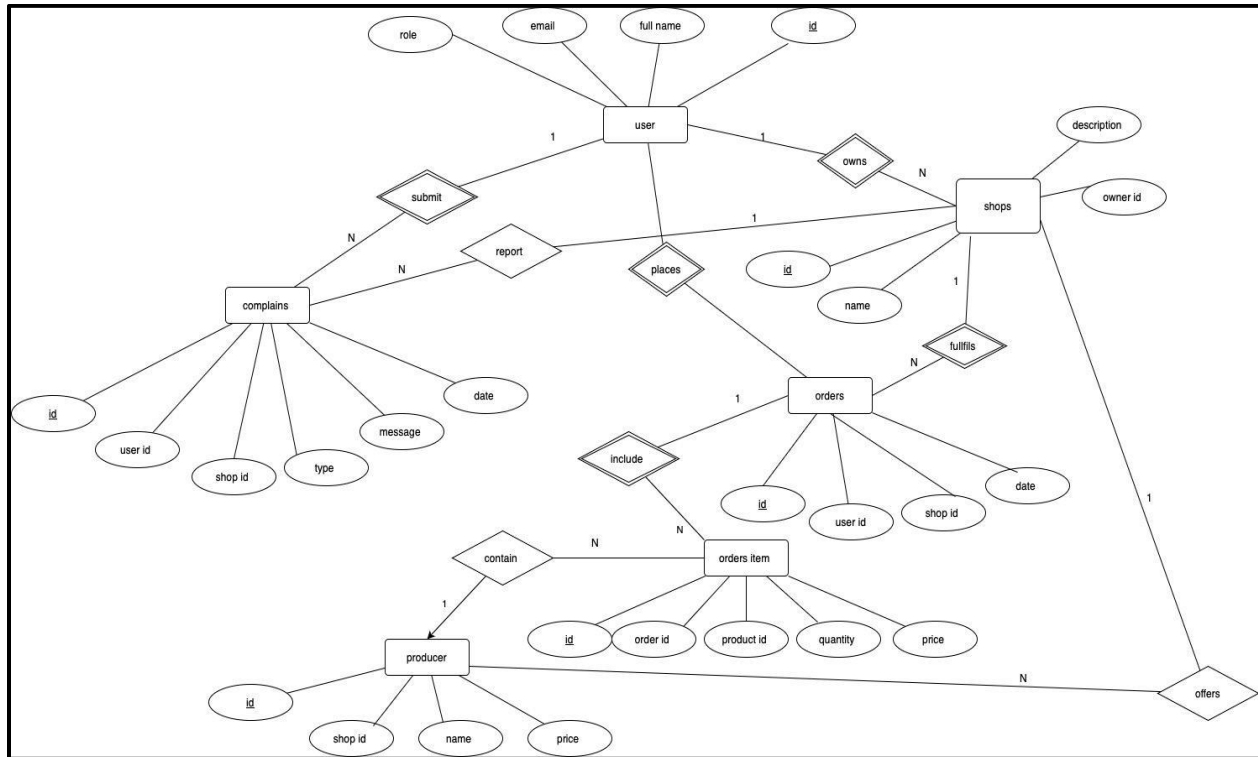


Figure 24: E-R Diagram

Figure 24 In spite of the fact that Shoply App utilizes the Firebase, a NoSQL document-based database, the taking after ERD is given to help in visualizing the structure and relationship of the information collections utilized .

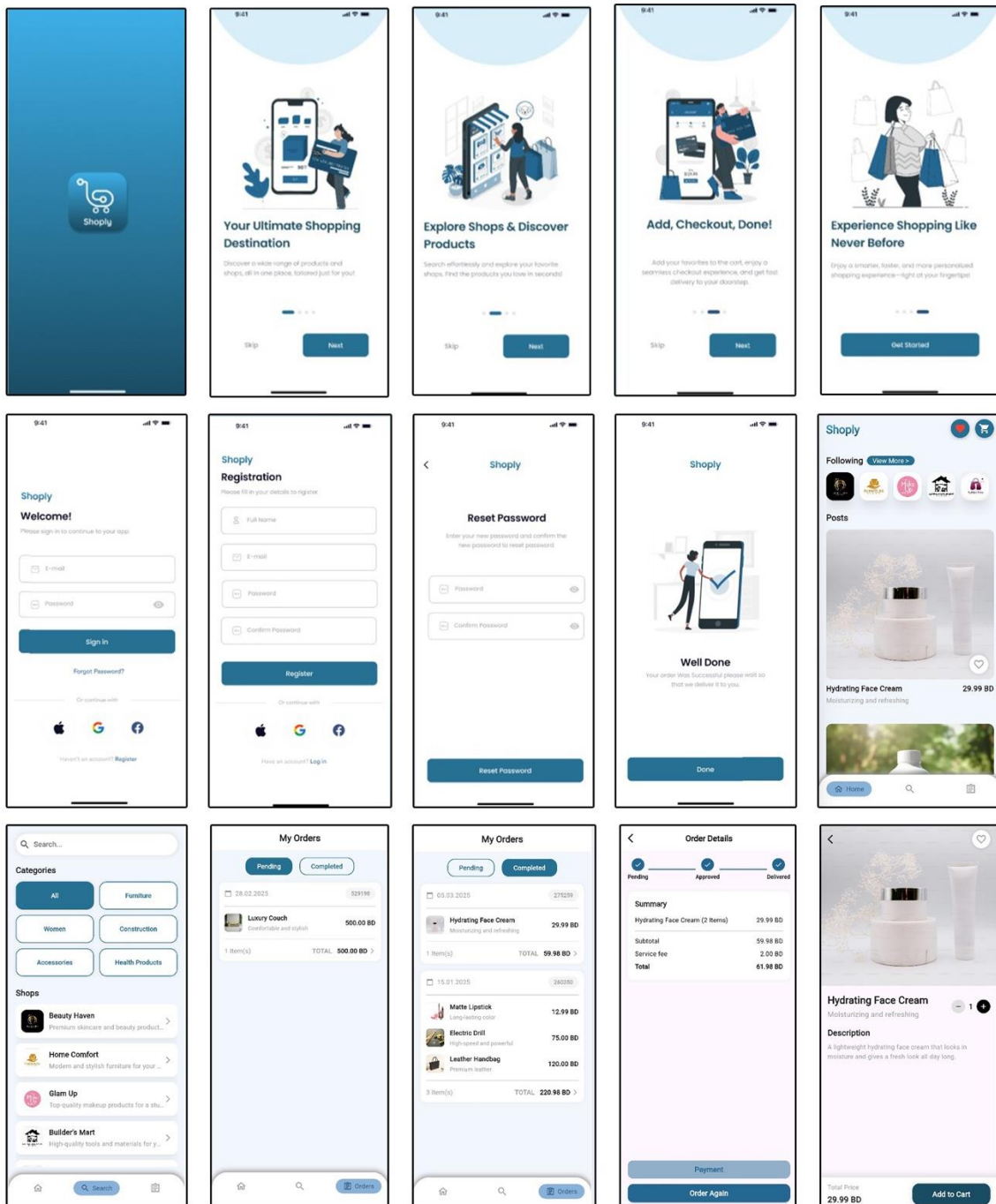
This ERD bridges Firebase's NoSQL dexterity and organized plan consideration—defining connections, get to ways, and scaling plans.

5.2 User Interface Design

Throughout development of the Shoply app, user interface (UI) design was thoughtfully addressed in a way that makes the experience smooth and intuitive to each user, be they customer or shop owner. The UI elements were carefully crafted to ensure easy navigation, neat visual grouping, and quick feedback on mobile as well as web channels.

The user interface focuses on usability and accessibility, making it easy for customers to navigate through products, operate shops, place items in carts, and make purchases. Shop owners also find it easy to manage products, upload images, and monitor orders. This strategy increases the general shopping experience while ensuring a simple and attractive user interface.

5.2.1 User Interface:



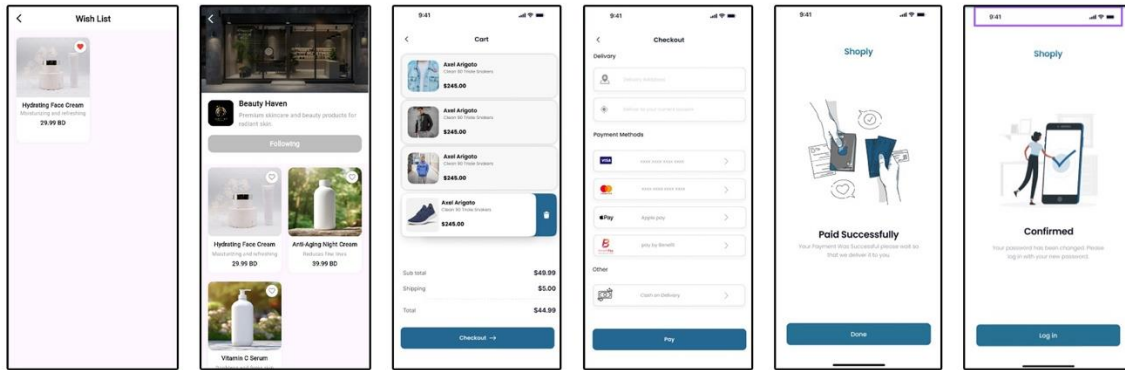


Figure 25: User Interface

The user interface of the Shoply App is user-friendly and minimalistic, letting users easily find their way to the home page where they can visit shops and shop through product groups. From this point, the users can find products, navigate shop pages, and add the products to the cart or Wishlist. The application also allows the users to sign up, login, and safely manage their individual profiles. Apart from that, it also provides facilities such as order tracking, safe payment via various methods, and successful payment notification. The site provides a seamless and interactive shopping experience optimized for local commerce.

5.2.2 Shop Owner Interface:

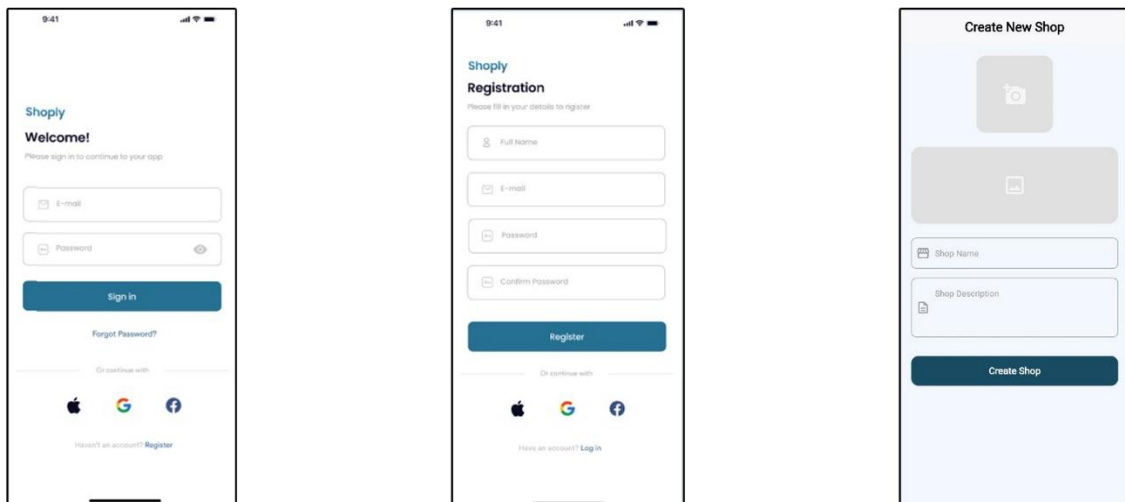


Figure 26: Shop Owner Interface

Shop owners interact with the Section for Interface that allows them to create new shop accounts while providing login abilities to start using the application. Through the Shop Owner interface, they can establish a new shop profile by performing essential steps that involve profile information entry and logo or cover image upload. Shop owners use this interface to manage their store content through product listings and pricing and updates for both their products and store description.

5.2.3 Admin Interface:

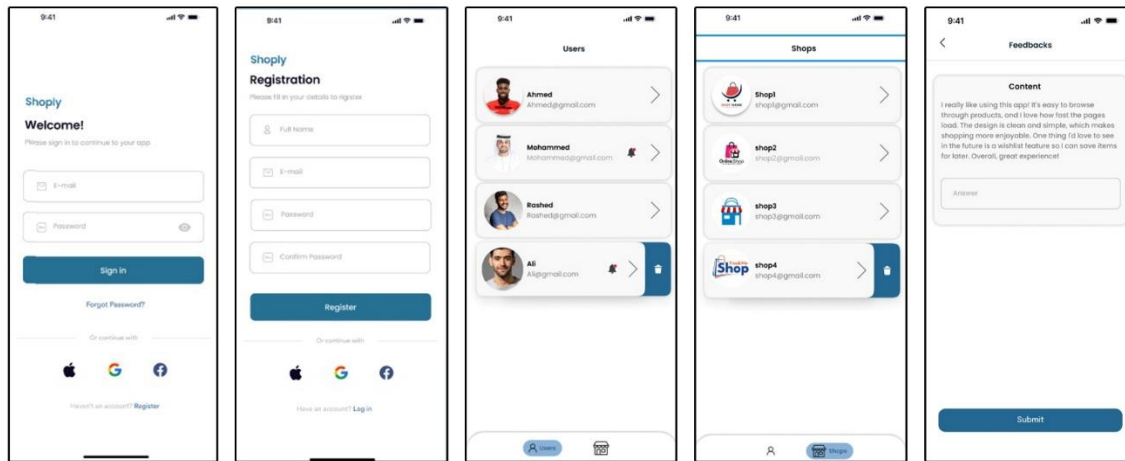


Figure 27: Admin Interface

Represents the Admin interface, by which administrators are registered and securely logged in to manage the platform. The admin can see and track registered users and use an active shops list via this interface. The admin can remove or deactivate accounts and shops when needed, so that the system remains kept in integrity and compliance. This interface provides needed tools to control platform activity and maintain overall system control. A feedback section included in the interface enables administrators to view customer and shop owner comments and provide appropriate responses thus enhancing service quality and establishing better relationships.

5.3 Functionality

This section presents the necessary functionalities developed in the Shoply application. All functions were designed to enhance user interaction, ease navigation, and provide needed services that agree with the entire aim of the app. Features of the system include product searching, filtering categories, and resetting passwords to offer personalized and safe shopping. These attributes not only support usability and legibility for the users but also help in maintaining system efficiency and reliability. The following is a concise description of each function with reference to its role in the system.

5.3.1 Search Function

```
12 class _SearchPageState extends State<SearchPage> {
13   String selectedCategory = 'All'; // Default selected category
14   String searchQuery = ''; // Search query state
15   final TextEditingController searchController = TextEditingController();
16
60   child: TextField(
61     controller: searchController,
62     onChanged: (value) {
63       setState(() {
64         searchQuery = value;
65       });
66     },
67     decoration: InputDecoration(
68       hintText: 'Search...',
69       prefixIcon: const Icon(Icons.search),
70       suffixIcon: searchQuery.isNotEmpty
71         ? GestureDetector(
72           onTap: () {
73             setState(() {
74               searchController.clear();
75               searchQuery = '';
76             });
77           },
78           child: const Icon(Icons.clear),
79         ) // GestureDetector
80       : null,
81     border: InputBorder.none,
82     contentPadding: const EdgeInsets.symmetric(vertical: 14),
83   ), // InputDecoration
84 ), // TextField
```

Figure 28: Search Function code

The above images show the implementation of the search feature in the app. From the first image, initial key variables such as `selectedCategory`, `searchQuery`, and the `searchController` are initialized to manage the user input as well as the category selection. The second image shows the `TextField` widget, from which users can enter search queries. As the user types, the `onChanged` property updates the query in real time. The category also includes a search icon and a clear icon to allow simple reset of the input so that there are no complications or hassles in searching for products.

5.3.2 Filter Category

```
130     boxShadow: [
131       BoxShadow(
132         color: Colors.grey.withOpacity(0.1),
133         blurRadius: 6,
134         spreadRadius: 2,
135       ), // BoxShadow
136     ],
137   ), // BoxDecoration
138   child: Center(
139     child: Text(
140       category,
141       style: TextStyle(
142         color: isSelected
143           ? Colors.white
144           : const Color(0xFF267093), // Text color
145         fontWeight: FontWeight.bold,
146       ), // TextStyle
147     ), // Text
148   ), // Center
149 ), // Container
150 ); // GestureDetector
151 },
152 ), // GridView.builder
```

```

95 // ✅ Category Buttons (Aligned)
96 GridView.builder(
97   shrinkWrap: true,
98   physics: const NeverScrollableScrollPhysics(),
99   itemCount: categories.length,
100   gridDelegate: const SliverGridDelegateWithFixedCrossAxisCount(
101     crossAxisCount: 2, // Two categories per row
102     crossAxisSpacing: 12,
103     mainAxisSpacing: 12,
104     childAspectRatio: 2.8,
105   ), // SliverGridDelegateWithFixedCrossAxisCount
106   itemBuilder: (context, index) {
107     final category = categories[index];
108     final bool isSelected = category == selectedCategory;
109
110     return GestureDetector(
111       onTap: () {
112         setState(() {
113           selectedCategory = category;
114         });
115       },
116       child: Container(
117         height: 40, // Reduced height
118         padding: const EdgeInsets.symmetric(
119           horizontal: 24,
120         ), // EdgeInsets.symmetric
121         decoration: BoxDecoration(
122           color: isSelected
123             ? const Color(0xFF267093) // Selected category color
124             : Colors.white,
125           borderRadius: BorderRadius.circular(16),
126           border: Border.all(
127             color: const Color(0xFF267093), // Border color
128             width: 1.5,
129           ), // Border.all

```

Figure 29: Filter Category code

The following images demonstrate the usage of the filter category functionality of the Shoply App. The first image shows how the user can select a specific category, and this updates the `selectedCategory` variable. This is triggered by interactive UI components reacting to user taps. The second image displays the logical filtering, in which products of the selected category only are displayed on the screen. This dynamic filtering helps customers eliminate items from their viewing area, hence making it easier to browse and personalize. Overall, the feature improves the consumer experience by making product discovery easier following consumer interest.

5.3.2 Reset Password

```
244     Align(
245       alignment: Alignment.centerRight,
246       child: TextButton(
247         onPressed: _sendResetLink,
248         child: const Text(
249           'Forgot Password?',
250           style: TextStyle(color: Colors.blueAccent),
251         ), // Text
252       ), // TextButton
253     ), // Align
80   Future<void> _sendResetLink() async {
81     final email = _emailController.text.trim();
82
83     if (email.isEmpty) {
84       setState(() {
85         _forgotError = 'Email is required';
86       });
87       return;
88     }
89
90     try {
91       // Optional: check if email exists in Firestore
92       final query = await FirebaseFirestore.instance
93         .collection('users')
94         .where('email', isEqualTo: email)
95         .get();
96
97       if (query.docs.isEmpty) {
98         setState(() {
99           _forgotError = 'Email does not exist';
100         });
101       }
102       return;
103     }
104     await FirebaseAuth.instance.sendPasswordResetEmail(email: email);
105
106     setState(() {
107       _forgotError = null;
108     });
109
110     ScaffoldMessenger.of(context).showSnackBar(
111       SnackBar(
112         content: Text('Reset link sent to $email'),
113         backgroundColor: Colors.green,
```

```

114         duration: const Duration(seconds: 3),
115     ), // Snackbar
116 );
117 } catch (e) {
118     print('Error sending reset email: $e');
119     setState(() {
120         _forgotError = 'Failed to send reset link. Try again.';
121     });
122 }
123 }

```

Figure 30: Reset Password code

The reset password functionality of Shoply App appears in the presented images. To start the password reset process users must enter their registered email that they previously used for account creation in the first image. The system sends a verification code to the specified email after receiving the email submission. The interface enables users to enter the received code and establish a new password. The third image reveals the system logic which confirms the code entry while performing secure updates to user credentials. Account security is strengthened through this complex procedure which promises users both safe restoration of access as well as strict protection measures.

In conclusion, the main features that have been implemented in the Shoply App—from product searching to resetting the password—are the fundamentals of an interactive and welcoming system. The features were designed with intuition to enhance the overall shopping experience, simplify the use interaction, and enable safe access to services. Each of the features plays a vital role towards addressing the app's main objectives of connecting customers to nearby shops effectively. The effective incorporation of these features echoes the project's emphasis on system reliability, performance, and usability.

Chapter 6

System Implementation and Testing

Synchronization of hardware and software components is a vital aspect in installing and testing the Shoply app. Following the development phase, careful and efficient coding was utilized to build a cross-platform mobile solution that supports both consumers and merchants within the local shopping ecosystem. Once the initial setup was completed, the app was put through testing to ensure that it functioned as required, enabling smooth and stable operation. Under this verification process, all the discrepancies, bugs, and performance issues were systematically identified and resolved. As a result, local users and businesses can trust Shoply as a safe and stable platform for secure digital shopping and business communication.

6.1 Tools and Languages Used

Shoply application was created by adopting a specially handpicked combination of programming languages, development frameworks, and cloud computing tools. Each was picked specifically to support effortless cross-platform functionality, secure data storage, and extensible user interface for both consumers as well as the shopkeepers. The sections below describe particularly utilized tools and technology in Shoply's implementation process and specifically the role it served and what the necessity in the larger scheme of the project.

This system was built using a combination of hardware and software resources including:

Software:

- **Flutter & Android Studio:** Flutter and Android Studio were utilized as the main development environments in developing the Shoply App. Android Studio facilitated a more enhanced virtual device environment, and the development team was able to code, debug, and test the application using various screen sizes and platforms during the development process.
- **Visual Studio Code:** Visual Studio Code was a major contributor to writing and maintaining the project codebase. With IntelliSense, integrated support with Git, and extension capabilities, it offered a lightweight and versatile alternative for rapid coding, especially in UI development.
- **Web Browser (e.g., Google Chrome):** The application was also thoroughly tested on Google Chrome, especially for the web version of Shoply. The use of the browser provided

cross-platform functionality and allowed real-time debugging and layout testing to improve responsiveness and performance.

- **Firestore:** Firestore served as the backend for the Shoply App and provided secure and scalable services such as authentication, Firestore database storage, and cloud hosting. It facilitated proper management of data and handling of users, which was a requirement for managing shop owners, customers, and products.
- **GitHub:** GitHub was utilized as the team collaboration and version control platform. It allowed team members to keep their code contributions, review pull requests, and have a well-structured and conflict-free codebase during the project cycle.
- **Communication App (e.g., WhatsApp):** WhatsApp was used as the primary tool of communication among team members. It facilitated coordination, daily updates, and immediate troubleshooting, hence the efficiency of team work and development.

Hardware:

- **Personal Laptops:** Shoply App development was carried out on personal high-performance laptops to ensure seamless system performance and efficient programming task execution. The laptops offered the capability to execute emulators, compile code, and test application features without any interruption. Their reliability ensured seamless progress throughout the project. Since the team trusted their own devices, the implementation team enjoyed flexibility, constant access to resources, and full control over the environment, which made the workflow significantly more efficient and productive at each phase of implementation.

6.1.1 Languages Utilized

In the development of the Shoply App, the following technologies and programming languages were employed:

- **Dart & Flutter:** The Shoply App development utilized Dart programming language together with Flutter framework as its core technologies. Dart provides a contemporary and efficient type-safe environment that Flutter uses for creating responsive cross-platform applications with visual consistency. Through using these technologies together the team was able to develop a seamless operating system which functioned on Android and iOS platforms as well as on the web.
- **CSS:** The web version application used Cascading Style Sheets (CSS) to improve its visual presentation. Through CSS complete design control of elements including layout structure and color theme alongside responsive functionality allowed the team to build an interface that would match Shoply's official branding.

- **JSON Files and API Integration:** JSON (JavaScript Object Notation) files were applied for formatted data transfer between application components, notably in dealing with product and user data. Secondly, integration through the API allowed interacting with Firebase services as well as external modules for integrating real-time synchronization of data as well as authentic and secure verification—features deemed essential for delivering users correct, responsive responses across the app.

6.2 Database/Cloud Firestore

For our project, we chose to use the Flutter framework and the development tools of Android Studio to build our app. The data storage capacities, which had to be strong, were taken care of by Firestore Database, which proved to be the core element responsible for storing and handling user information, shop data, product data, and images with maximum security and efficiency. Although the integration of chat functionality was a huge challenge initially, the process of integration with Firebase was totally seamless and stable, and this worked wonders for the overall success of our Shoply App. The real-time updates of Firestore assisted in smooth synchronization among users, and hence the system always remained responsive. Here are a sequence of well-made screenshots showing the structure and arrangement of our database in the Firestore Database:

6.2.1 User table:

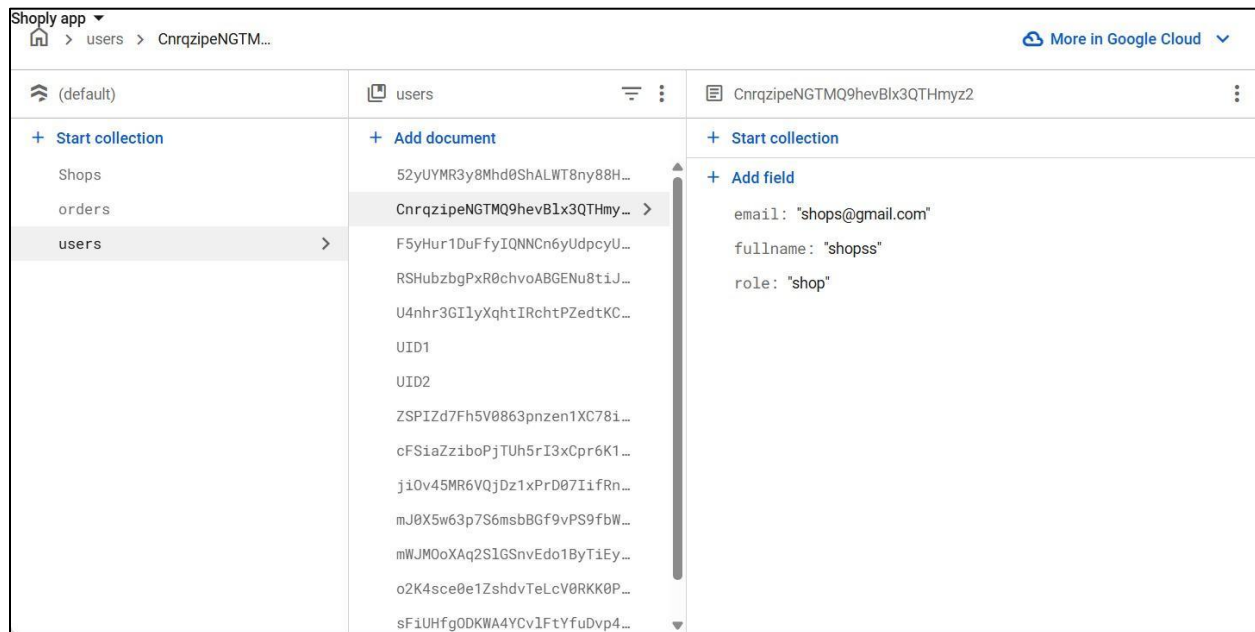


Figure 31: User Table

6.2.2 Orders table:

Shoply app More in Google Cloud		
orders QbDeoD8ziFRtV...		
(default)	orders	QbDeoD8ziFRtVha1wpNH
+ Start collection	+ Add document	+ Start collection
Shops	QbDeoD8ziFRtVha1wpNH	items
orders		+ Add field
users		date: May 2, 2025 at 12:00:00 AM UTC+3
		serialNumber: 12345
		status: "Pending"
		userId: "52yUYMR3y8Mhd0ShALWT8ny88HE2"

Figure 32: Orders Table

6.2.3 Shops table:

Shoply app More in Google Cloud		
Shops vMzwzk2NGe0...		
(default)	Shops	vMzwzk2NGe0mrisN4803
+ Start collection	+ Add document	+ Start collection
Shops	vMzwzk2NGe0mrisN4803	products
orders		+ Add field
users		backgroundImage: "assets/shopbackground1.png"
		description: "Premium skincare and beauty products for radiant skin."
		isFollowed: false
		logo: "assets/shop1.png"
		name: "Beauty Haven"

Figure 33: Shops Table

6.2.4 Items table:

Shoply app More in Google Cloud		
orders > QbDeoD8ziFRtVha1wpNH	items	Im2MHibBgPgO3GxMoqsd
+ Start collection	+ Add document	+ Start collection
items >	Im2MHibBgPgO3GxMoqsd >	+ Add field
+ Add field date: May 2, 2025 at 12:00:00 AM UTC+3 serialNumber: 12345 status: "Pending" userId: "52yUYMR3y8Mhd0ShALWT8n"		image: "https://www.google.com/url?sa=i&url=https%3A%2F%2Fbahrain.whizzcart.com%2Fproduct%2F187-cordless-drill-set-12v-power-drill-set-with-battery-and-charger-electric-drill-driver-drill-bits-3-8-keyless-chuck-21-1-torque-setting-180-inch-lbs-with-led-electric-drill-set-yellow%2F&psig=AOvVaw215B8RANI8x-C8PSdLpO52&ust=1746278339274000&source=images&cd=vfe&op"
		longDescription: "" name: "Electric Drill" price: 75 quantity: 1 shortDescription: "High-speed and powerful"

Figure 34: Items Table

6.2.5 Products table:

Shoply app More in Google Cloud		
Shops > vMzwzk2NGe0mrisN4803	products	ThUIZusg0HSwxqKD3B4r
+ Start collection	+ Add document	+ Start collection
products >	ThUIZusg0HSwxqKD3B4r >	+ Add field
+ Add field backgroundImage: "assets/shopbackgr"		category: "Accessories"
description: "Premium skincare and beauty products for radiant skin."		image: "https://www.google.com/imgres?q=makeup%20brush%20set&imgurl=https%3A%2F%2Fimages-na.ssl-images-amazon.com%2Fimages%2F%2F1dpq-%2B%2Bv1L.jpg&imgrefurl=https%3A%2F%2Fbahrain.whizzcart.com%2Fproduct%2F187-cordless-drill-set-12v-power-drill-set-with-battery-and-charger-electric-drill-driver-drill-bits-3-8-keyless-chuck-21-1-torque-setting-180-inch-lbs-with-led-electric-drill-set-yellow%2F&psig=AOvVaw215B8RANI8x-C8PSdLpO52&ust=1746278339274000&source=images&cd=vfe&op"
isFollowed: false		isInCart: false isWishlisted: false
logo: "assets/shop1.png"		longDescription: "A set of high-quality makeup brushes..."
name: "Beauty Haven"		name: "Makeup Brush Set"
		price: "19 99"

Figure 35: Products Table

6.3 Software Testing

In developing our Shoply application, we carried out extensive testing of the software comprising a comprehensive evaluation under diverse situations to ascertain their success or failure prospects. We predominantly sought to identify and eliminate any bugs or functionality issues that could hinder the system's performance before going live. This level of attention to detail allowed us to improve the overall usability experience. By testing early, we were able to detect and resolve issues to enable smooth functionality on each feature. Hence, the users can navigate and use the app with assurance that the system is reliable and efficient. It, however, needs to be understood that although it is always good to minimize all errors, it remains more of an ideal than a certainty to attain absolute perfection.

6.3.1 Integration/E2E Tests (Front-End)

On the client side, we used Flutter's `integration_test` package to test actual user flows on Android and iOS simulators. Automated tests open the Shoply application, log in with valid credentials, browse through the list of products, add products to cart, and perform a mock checkout. We measure each screen switch to make sure it finishes in under five seconds, and we also check that cart, payment type, and thank-you page show up as we want them to. We loaded these tests 50 in a row on baseline emulator configurations, and there weren't any UI timeouts or crashes. That confirmed that our UI state management and navigation are quite solid in the real world.

6.3.2 Integration Tests with Firebase Emulator (Back-End)

We tested our back-end with the Firebase Emulator Suite so that it is like it's real. We had emulators running for Firestore, Auth and Cloud Functions locally so nothing in production gets accessed. Our test scripts created mock users in Auth, pushed dummy orders to Firestore, then called the Cloud Functions as they would in real life. Then we checked if the stock amounts were declining in the correct manner, if the received order documents were updated with proper status, and there were no "permission denied" sorts of errors. We ran more than 100 test cases, some were corner cases, but none of them were a failure. That makes us confident that our back-end logic as well as security rules are appropriate before we go live.

Test Case ID	Layer	Test Scenario	Test Steps	Expected Results	Actual Results	Pass/Fail
1	Back-End	Valid Order Processing	Start emulators; create test user; write valid order doc; trigger Cloud Function	Function completes; stock decremented; order status 'processed'	As expected	Pass
2	Back-End	Reject Missing Fields	Write order doc missing shop_id; trigger Cloud Function	Function returns 400 error; no stock change	As expected	Pass
3	Back-End	High-Volume Order Burst	Loop write 50 orders rapidly; trigger Function for each	All functions run <200ms; no throttling errors	All under threshold	Pass
4	Back-End	Security-Rules Enforcement	Unauthenticated read of another user's orders; update someone else's shop doc	Both operations denied with permission_denied	Both denied	Pass
5	Back-End	Error-Path in Notification Function	Create order doc with invalid notifyEmail; trigger notification Function	Function logs error; does not crash	Error handled	Pass
6	Front-End	User Registration Flow	Launch app; tap Sign Up; enter valid data; submit form	User created; redirected to home	As expected	Pass
7	Front-End	Login & Logout	Launch app; enter credentials; sign in; tap Logout	Login succeeds; logout returns to login screen	As expected	Pass
8	Front-End	Search & Filter Products	Login; search 'Shoes'; apply 'Sports' category	Only sports shoes appear; list scrolls smoothly	As expected	Pass
9	Front-End	Wishlist and Cart Persistence	Login; add product A to Wishlist; add product B to Cart; restart app	Wishlist contains A; Cart contains B after restart	As expected	Pass
10	Front-End	Password Reset Integration	Tap Forgot Password; enter email; fetch emulator code; enter code & new password; login	Password reset succeeds; new password login works	As expected	Pass

Table 18: Software testing

6.3.2 Login Validation Testing

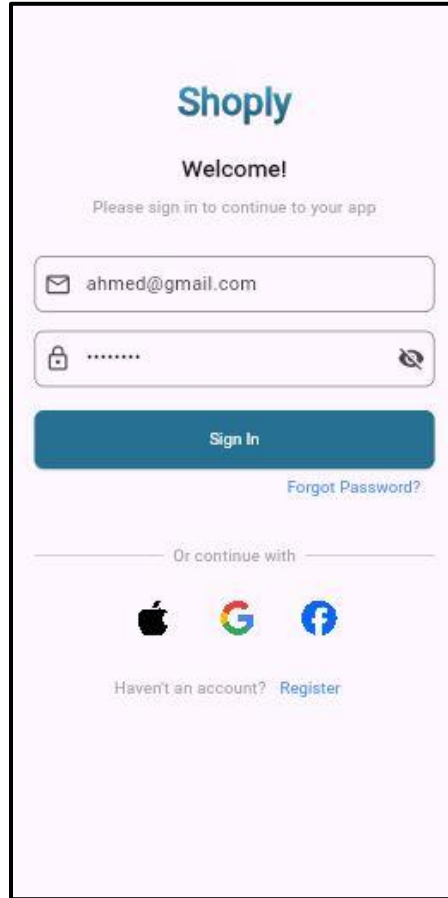


Figure 36: Login Validation Testing

The login functionality in the Shoply application performed correctly. The user initiates by entering his registered email address and password and proceeds by clicking on the Sign In button. Upon confirmation, the access is granted and, accordingly, the user is navigated to the home screen, where such crucial features as the product feed, wishlist, cart, and navigation through the shop are provided. Login was fast and seamless, confirming that the authentication process works well and securely under normal usage scenarios.

6.3.3 Register Validation Testing

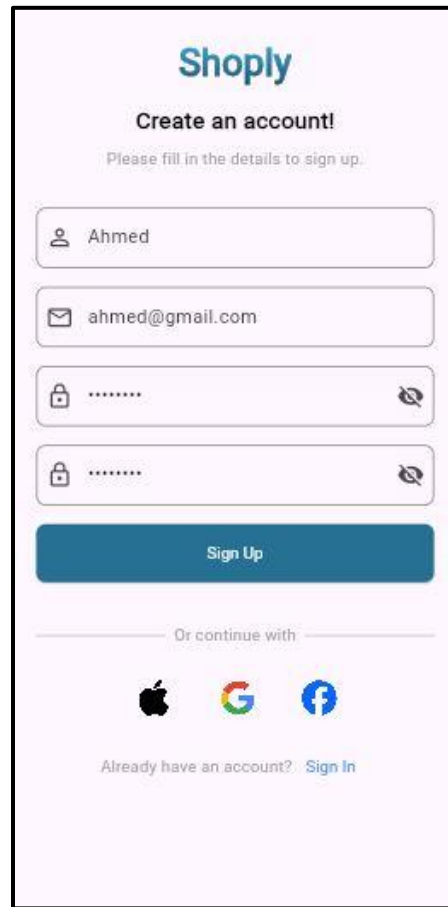
The image shows a mobile application registration screen for 'Shoply'. At the top, the 'Shoply' logo is displayed in blue. Below it, the text 'Create an account!' is centered, followed by a smaller instruction: 'Please fill in the details to sign up.' The form consists of four input fields: a name field containing 'Ahmed', an email field containing 'ahmed@gmail.com', a password field with masked characters '*****' and a toggle icon, and a re-type password field also with masked characters and a toggle icon. A prominent blue 'Sign Up' button is located below the password fields. Underneath the button, a horizontal line separates it from the text 'Or continue with', which is followed by three social media icons: Apple, Google, and Facebook. At the bottom of the screen, the text 'Already have an account?' is followed by a blue link labeled 'Sign In'.

Figure 37: Register Validation Testing

The image shows that the register function was appropriately tested on the Shoply application. The full name, email, password, and re-typed password of the new user are requested to be filled out. The entered input in the form is stored by the system after the button Sign Up has been clicked. This action renders the user accessing the core app features such as shopping, wishlisting, and ordering tracking right away. The process was tried for facilitation of smooth use, safe data handling, and applicable redirection upon a user signing up.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

In conclusion, this report presents the successful achievement of Shoply application development as a digital marketplace that links Bahrain-based small businesses to customers through its mobile-friendly platform for marketplace interactions. The major objective was to enhance business visibility together with improving product exploration while providing an accessible system for customer interactions. The development of this application prioritized usability along with user-friendly navigation and the vital components that included product listings and real-time order tracking and customer-to-shop messaging features.

The project maintained its programmatic development plan during every phase to deliver successful progress and expedite completion milestones. The project utilized effective diagrams and models to explain workflows and architecture and components' interactions thus resulting in improved design and implementation activities. Testing involved multiple scenarios which checked for functional quality along with efficiency and user-friendly interfaces as well as deployment steps to finalize the application. The current Shoply version satisfies its main goals but ongoing work plans to focus on developing enhanced analytics features and simplifying checkout payments within the application.

7.2 Future Work

While our program fulfills the most basic necessities using a sequence of functions, we have a few extra items in store for which it can provide. There were some functionalities we wanted to implement but were unable to because of the time constraint for developing this project along with other academic responsibilities taking place simultaneously. The most important areas to focus on in the future are:

There are some functionalities and things we want to add in the future:

1. Introducing a wishlist feature that allows users to save favorite products for future purchases.
2. Easy of sending alerts and notices for order status, promotions, and customer-shop activity.
3. Implementing the capability of search inside the app based on a given shop, product, or category.
4. Adding tracking features for a product such as live order status and delivery monitoring.

5. Creating special promotions and seasonal discount devoted page from featuring stores.
6. Enabling users to rate their shopping experience and provide feedback to improve service quality.

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